Computer Law & Security Review 55 (2024) 106056

Contents lists available at ScienceDirect

Computer Law & Security Review: The International Journal of Technology Law and Practice

journal homepage: www.elsevier.com/locate/clsr

Copyright protection during the training stage of generative AI: Industry-oriented U.S. law, rights-oriented EU law, and fair remuneration rights for generative AI training under the UN's international governance regime for AI

Kaigeng Li, Hong Wu *, Yupeng Dong

NingboTech University, Ningbo, Zhejiang, China

ARTICLE INFO

Keywords:
Generative AI
Copyright
Training
Data content
Fair use
Comparative study

ABSTRACT

Generative AI relies on simulating and learning from complex data distributions to automatically generate new, meaningful content from large datasets. Training generative AI models carries the risk of copyright infringement. How can we balance the development of generative AI technology with copyright protection during the training stage of these models? In contrast to existing legal scholarship, this article conducts a comparative study of industry-oriented U.S. copyright law and rights-oriented EU copyright law. It draws insights from the concepts of the tragedy of the commons and the tragedy of the anti-commons, proposing the creation of fair remuneration rights for Generative AI training under the UN's international governance regime for AI. This article offers a typological analysis of potential operational models for fair remuneration rights in Generative AI training, drawing analogies to existing remuneration rights in international copyright treaties. The goal is to provide an open framework for further discussion within the international academic community.

1. Introduction

Generative AI (GenAI), which can create text, images, and other forms of content, provides unique opportunities for innovation.[1] However, as highlighted by recent judicial lawsuits, it also presents significant challenges for artists, creators, and publishers regarding how their creative content is produced, distributed, used, and consumed. In the US, the Associated Press reached a two-year agreement with OpenAI in July 2023, allowing OpenAI to use portions of its text archive since 1985 to help train ChatGPT.[2] Later, in December 2023, Axel Springer signed an agreement with OpenAI, stipulating that OpenAI will pay to use content from Axel Springer's publications to enhance ChatGPT's responses and train its AI tools.[3] However, the New York Times was unable to reach a cooperation agreement with OpenAI and filed a lawsuit against OpenAI on December 27, 2023.[4] The lawsuit alleges that OpenAI used millions of articles from the New York Times to train its ChatGPT

This article is funded by the 2021 National Social Science Foundation Project (21BXW040).

* Corresponding author.
E-mail addresses: likaigeng@nbt.edu.cn (K. Li), mllxfhw2@zju.edu.cn (H. Wu), yupengdong@vip.126.com (Y. Dong).

[1] Tojin T Eapen and others, 'How Generative AI Can Augment Human Creativity' [2023] *Harvard Business Review*.
[2] Matt O' Brien, 'ChatGPT-Maker OpenAI Signs Deal with AP to License News Stories' (*APnews.com*, 13 July 2023).
[3] OpenAI, 'Partnership with Axel Springer to Deepen Beneficial Use of AI in Journalism' (*OpenAI.com*, 13 December 2023).
[4] *The New York Times Company v Microsoft Corporation,* 1:23-cv-11195, (SDNY).

https://doi.org/10.1016/j.clsr.2024.106056

Available online 1 October 2024
2212-473X/©2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

and other GPT series-based large language models without authorization, thereby infringing on its copyrights. OpenAI chose not to disclose specific details about the training datasets, model architecture, hardware configurations, training computation processes, dataset construction methods, or training strategies. This lack of transparency makes it difficult for external parties to ascertain the exact nature of the infringement.[5] Additionally, in January 2023, Sarah Andersen, Kelly McKernan, and Karla Ortiz sued Stability AI, Midjourney, DeviantArt, and Runway AI.[6] They claimed that the companies used their works without permission to train an AI-based software library and model, which then generated new images. These companies are accused of directly infringing the artists' copyrights, as well as indirect copyright infringement and other violations. In May 2024, U.S. District Judge William Orrick stated that these lawsuits might have sufficient grounds to proceed and should be examined in the next phase of the case under various theories and precedents of direct and inducement infringement under copyright law.[7] Across the Atlantic, in March 2024, the French Competition Authority (FCA) fined Google €250 million for failing to keep its pledge to broker fair agreements with media outlets for publishing links to their content. Google was also accused of using press articles to train its AI technology without informing the media outlets.[8]

The risk of copyright infringement can arise at various phases of GenAI model development.[9] During the data collection phase, the unrestricted use of training data content created through significant investment by copyright holders may constitute an infringement of reproduction rights. In the data pre-processing phase, activities such as cleaning, standardizing, annotating, and feature extraction of the collected data take place. During training, the model learns from the quality content of the copyright holders, potentially reflecting similar quality and information in the generated content, thus posing a risk of infringing on adaptation rights and other rights. In the content generation phase, the GenAI might produce fragments, detailed summaries, or even verbatim excerpts identical or highly similar to copyrighted works. This could effectively deprive copyright holders of their rightful revenue, posing risks of infringing on the rights of reproduction, adaptation, and public communication.

International and national copyright laws must actively tackle the challenge of balancing the development of GenAI technology with the protection of copyright holders' interests, especially during the training stage of GenAI. It is evident that the United Nations (UN) and the World Intellectual Property Organization Copyright Treaty (the WIPO Copyright Treaty)[10] have increasingly focused on copyright issues related to GenAI model training in recent years. However, no international agreement has yet specifically addressed this issue, and discussions remain ongoing. As a result, it is necessary to analyze the existing international copyright legal framework to lay the groundwork for exploring feasible solutions in the future. This article examines the limits on protection outlined in these international agreements to assess their applicability to GenAI model training. The approach to limitations and exceptions established by the Berne Convention for the Protection of Literary and Artistic Works (the Berne Convention)[11] grants countries significant legislative freedom, allowing them considerable latitude in establishing specific limitations and exceptions to copyright protection. However, this flexibility also introduces notable challenges, particularly the lack of consensus on the rules governing copyright limitations and exceptions across different countries. Consequently, substantial differences exist in how these issues are addressed, leading to varying legal environments for GenAI model training and copyright protection worldwide. U.S. law currently addresses the issue of copyright protection for data content in the GenAI training stage through the application of case law. Overall, U.S. rules are industry-oriented. The concept of non-expressive fair use is adopted in cases involving technologies with a certain degree of similarity to GenAI, such as reverse engineering, search engines, plagiarism detection software, and book digitization.[12] In terms of training data disclosure, the U.S. currently has no formal mandatory regulations. However, on April 9, 2024, Congressman Adam Schiff introduced a new bill named the GenAI Copyright Disclosure Act,[13] which requires AI companies to disclose the use of copyrighted training data content. On the other side of the Atlantic, EU law addresses this issue through the Digital Single Market Directive (the DSM Directive)[14] and the Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (the AI Act).[15] Unlike U.S. law, EU law tends to be rights-oriented. The EU generally does not allow AI model providers to use copyrighted content for training data unconditionally and for free, with limited exception or limitation. The DSM Directive has introduced Article 4, which allows exception or limitation for the reproduction and extraction of works or other subject matter for text and data mining under certain conditions. Rightholders can choose to explicitly reserve their rights to works or other subject matter to prevent text and data mining in the AI training stage, unless it is for scientific research purposes. In addition, Article 53 of the AI Act requires AI model providers to draft and publicly disclose sufficiently detailed summaries of the content used to train AI models, based on templates provided by the AI Office.

The legal academia have already begun investigating the issue of copyright protection for data content during the training stage of GenAI models. Some scholars believe that training GenAI models carries the risk of copyright infringement. Gil Appel, Juliana Neelbauer, and David A. Schweidel have noted that GenAI faces significant intellectual property issues.[16] In the context of text-to-image models, Professor Matthew Sag identifies some specific situations where memorization of training data is more likely, which increases the likelihood of infringement.[17] Regarding whether using copyrighted works constitutes infringement, some scholars support the idea that using copyrighted works for data training by GenAI can be considered fair use and, therefore, does not infringe on copyright. Mark Lemley and Bryan Casey argue that training machine learning models on copyrighted material should generally be

[5] Audrey Pope, 'NYT v. OpenAI: The Times's About-Face' (*Harvard Law Review*, 10 April 2024).

[6] *Anderson v Stability AI Ltd,* No 23-CV-00201-WHO, 2023 WL 7132064 (ND Cal Oct 20, 2023).

[7] Blake Brittain, 'Stability AI, Midjourney Should Face Artists' Copyright Case, Judge Says' (*Reuters*, 9 May 2024).

[8] Reuters, 'French Competition Watchdog Hits Google with 250 Million Euro Fine' (*Reuters.com*, 20 March 2024).

[9] King and Wood Mallensons, 'Risks of Gen AI - Safeguarding Intellectual Property in the AI Era' (*Lexology*, 12 June 2024).

[10] World Intellectual Property Organization Copyright Treaty, December 20, 1996.

[11] Berne Convention for the Protection of Literary and Artistic Works, September 9, 1886, as amended on September 28, 1979.

[12] *Sega Enters v Accolade, Inc,* 977 F2d 1510, 1514 (9th Cir 1992); *Sony Computer Ent v Connectix Corp,* 2203 F3d 596, 608 (9th Cir 2000); *Kelly v Arriba Soft Corp,* 336 F3d 811 (9th Cir 2003); *AV ex rel Vanderhye v iParadigms, LLC,* 562 F3d 630, 644-45 (4th Cir 2009).

[13] Adam Schiff, 'Rep. Schiff Introduces Groundbreaking Bill to Create AI Transparency between Creators and Companies' (*Schiff.house.gov*, 9 April 2024).

[14] Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on Copyright and Related Rights in the Digital Single Market and Amending Directives 96/9/EC and 2001/29/EC.

[15] Regulation of the European Parliament and of the Council Laying Down Harmonized Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts, 2021/0106 (COD).

[16] Gil Appel, Juliana Neelbauer and David A Schweidel, 'Generative AI Has an Intellectual Property Problem' (*Harvard Business Review*, 7 April 2023).

[17] Matthew Sag, 'Copyright Safety for Generative AI' (2023) 61 Houston Law Review 295.

permitted under fair use, as long as the model's outputs go beyond the original purpose of the inputs.[18] Jessica Gillotte argues that fair use would serve as a valid defense because training an AI model transforms the original work and is unlikely to damage the original artist's market for the copyrighted work.[19] Conversely, some scholars hold the opposite view. Jacob Alhadeff, Cooper Cuene, and Max Del Real argue that even if text-to-image models are not deemed transformative, the potential for market usurpation alone is sufficient to negate fair use.[20] In contrast to these two viewpoints, Professor Benjamin L. W. Sobel believes that the current fair use doctrine in U.S. copyright law is in crisis when addressing copyright issues related to GenAI. He argues that if U.S. courts reject the fair use defense for machine learning, valuable innovation may move to other jurisdictions or halt entirely. Conversely, if courts find the technology to be fair use, sophisticated software may divert the rightful earnings from the authors of input data.[21] As a solution, some scholars propose introducing a statutory license for GenAI training purposes into EU law. This proposal aims to provide remuneration to rightholders based on the rights to science and culture, freedom of artistic expression, and the protection of creators' moral and material interests derived from their scientific, literary, or artistic works.[22]

In contrast to existing legal scholarship, this article argues, through a comparative study of U.S. and EU law, that insights can be drawn from the concepts of the tragedy of the commons[23] and the tragedy of the anti-commons[24] to adopt a more balanced and effective approach in addressing the copyright challenges of GenAI data training. It argues that the industry-oriented approach of U.S. law fails to adequately consider economic compensation for copyright holders, potentially exacerbating tensions within the GenAI industry. Conversely, the rights-oriented approach of EU law, which allows copyright holders to make special reservations to prevent their works from being used for GenAI data training, may lead to issues such as misalignment with the vast data demands, as well as data silos and model bias. Given that copyright holders have incentives to exercise the right to reservation under rights-oriented EU copyright law, the exercise of this right fundamentally does not differ from the "prior consent" model, where the use of copyrighted works requires prior consent from the copyright holder. The copyright transaction model of "right to reservation, prior consent, and royalties" cannot meet the massive data learning needs of the GenAI era. This conflict easily leads to low transaction efficiency, increased transaction costs, and ineffective GenAI data training. Additionally, the inability to access comprehensive and diverse datasets during training can affect the fairness and effectiveness of the models. In the face of the copyright challenges posed by the development of GenAI models, national legislation alone is insufficient and does not support the widespread global application of GenAI technology. It is therefore ideal for different jurisdictions to collaborate and address these challenges by formulating binding or non-binding norms at the global level. This article proposes the creation of a pathway within the UN-led international governance regime for AI to balance GenAI model training with copyright protection. Regarding the institution(s) specifically responsible for developing guidelines, norms, and rules to balance GenAI model training with copyright protection, WIPO appears to be a suitable choice. However, given WIPO's tendency to move slowly, this article does not foresee binding norms for balancing GenAI innovation and copyright protection being agreed upon among member states in the near future. A more practical approach might be to first build a broad consensus on the issue of GenAI model training and copyright protection through the annual WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI). Once a broad consensus is achieved, norms—whether non-binding or binding—could then be proposed, depending on the political dynamics among countries. Drawing on insights from the tragedy of the commons and the tragedy of the anticommons, one viable solution to address the shortcomings of both U.S. industry-oriented copyright law and EU rights-oriented copyright law in balancing the development of GenAI technology with copyright protection is to provide fair remuneration to copyright holders whose works are used without compensation in GenAI model training. This article offers a typological analysis of potential operational models for the right to fair remuneration for GenAI training, drawing analogies to existing types of remuneration rights in international copyright treaties. The goal is to provide an open framework for further discussion within the international academic community. Moreover, while there are precedents for international treaties mandating the global implementation of uniform copyright exceptions—such as the Marrakesh Treaty to Facilitate Access to Published Works for Persons Who Are Blind, Visually Impaired, or Otherwise Print Disabled (the Marrakesh Treaty)—this article suggests that, given the complex political environment surrounding copyright exceptions in the current international copyright law landscape, it may be more practical to continue leaving the determination of copyright exceptions for GenAI model training to national legislatures and courts.

In addressing the copyright issues during the training stage of GenAI, this article is divided into four parts. Section 2 describes the technical dependency of GenAI on training data and the associated copyright infringement risks. Section 3 analyzes the international and national legal frameworks for copyright protection related to GenAI data training. Section 4 highlights copyright issues through a comparative analysis of U.S. and EU law and proposes cooperation among jurisdictions to address these challenges. Section 5 concludes the article.

2. Dependency of GenAI on training data and copyright infringement risks

2.1. Dependency of GenAI on training data

GenAI relies on simulating and learning from complex data distributions to automatically generate new, meaningful content from large datasets. The core of GenAI lies in its innovativeness and creativity, enabling it to not only understand and interpret existing data patterns but also transcend these known patterns to create unprecedented outputs. This capability has shown vast application prospects in various fields, including natural language processing, computer vision, music composition, and art design. The basic architecture of GenAI is typically built on deep learning, particularly models such as Variational Autoencoders (VAEs),[25] Generative Adversarial Networks (GANs),[26] and Transformers. These models aim to learn the distribution of latent data spaces and generate new data by sampling from these distributions. For instance, GANs consist of two components: the generator and the discriminator. These components compete with each other to optimize the quality of the generated data, making it increasingly close to the real data distribution.[27] VAEs, on the other hand, learn the probabilistic

[18] Mark A Lemley and Bryan Casey, 'Fair Learning' (2021) 99 Texas Law Review 743.

[19] Jessica Gillotte, 'Copyright Infringement in AI-Generated Artworks' (2020) 53 UC Davis Law Review 2655.

[20] Jacob Alhadeff, Cooper Cuene and Max Del Real, 'Limits of Algorithmic Fair Use' (2024) 19 Washington Journal of Law, Technology & Arts 1.

[21] Benjamin LW Sobel, 'Artificial Intelligence's Fair Use Crisis' (2017) 41 Columbia Journal of Law & The Arts 45.

[22] Christophe Geiger and Vincenzo Iaia, 'The Forgotten Creator: Towards a Statutory Remuneration Right for Machine Learning of Generative AI' (2024) 52 Computer Law & Security Review.

[23] G Hardin, 'The Tragedy of the Commons' (1968) 162 Science 1243.

[24] MA Heller, 'The Tragedy of the Anticommons: Property in the Transition from Marx to Markets' (1998) 111 Harvard Law Review 621.

[25] Diederik P Kingma and Max Welling, 'An Introduction to Variational Autoencoders' (2019) 12 Foundations and Trends in Machine Learning 307.

[26] Kunfeng Wang and others, 'Generative Adversarial Networks: Introduction and Outlook' (2017) 4 IEEE/CAA Journal of Automatica Sinica 588.

[27] ibid.

distribution of the data through an encoder-decoder structure, which allows them to reconstruct and create new samples based on this learned distribution.[28] As a core branch of deep learning technology, the development and advancement of GenAI largely depend on the extensive, diverse, and high-quality training data it relies on.[29] On one hand, training data provides the foundational knowledge for GenAI. Whether it's a text generation model in natural language processing or an image generation model in computer vision, these models learn patterns and rules from large-scale datasets to understand different representations of the world.[30] For instance, in text generation tasks, the model analyzes vast amounts of text fragments to understand grammatical structures, lexical collocations, and contextual logic.[31] In image generation tasks, the model learns the organization of visual elements such as color, shape, and texture by parsing countless combinations of pixels.[32] On the other hand, the scale of the data determines the extent to which the model can capture the complexities of the real world. Large-scale datasets help models effectively reduce overfitting, allowing them to generalize well even when faced with previously unseen data.[33] For example, when training an AI model to generate human face images, if the dataset includes photos of faces of different genders, ages, races, and expressions, the model can learn these features and generate new, realistic human faces after sufficient training.[34]

Computer science literature has employed various "extraction attacks" to demonstrate that, in some cases, GenAI may memorize details from some works in the training data, although these instances of memorization represent only a tiny proportion of the overall training datasets.[35] For instance, Nicholas Carlini and his colleagues discovered that out of the 175 million potential copies, 109 could reasonably be considered "near-copies" of images in the training data.[36] Their research indicates that the extent of memorization varies based on the design of different GenAI models. Memorization is more likely under the following conditions: when models are trained on numerous duplicates of the same work, when images are associated with unique text descriptions, and when the ratio of the model size to the training data is relatively large.[37]

Moreover, with the development of adversarial learning and self-supervised learning, GenAI is not limited to static training data. It can achieve self-iteration and continuous optimization through mechanisms such as self-play or contrastive learning, further expanding the model's knowledge boundaries and innovative capabilities.[38] Therefore, the essence of GenAI is its data-driven nature, with its core competitiveness stemming from the deep mining and efficient utilization of training data. Without data training, GenAI would lose its lifeline, unable to understand and reproduce the diversity and complexity of the real world, and unable to demonstrate its powerful creativity and practicality. Only with sufficiently rich and high-quality data can GenAI truly fulfill its potential.

2.2. Copyright infringement risks in GenAI data training

Training GenAI models carries the risk of copyright infringement. If memorization during the GenAI data training stage is technically possible,[39] then copyright infringement is also a possibility. The training data for GenAI typically consists of large amounts of data content, broadly categorized into three sources. Besides content from the public domain and content legally authorized through contractual agreements, the third category involves unauthorized content protected by copyright. Methods of obtaining such data often include using web scraping technology to collect online data, illegally accessing database content, and digitizing non-electronic data without permission. For instance, developing text-to-image GenAIs like Stable Diffusion involves training models with extensive datasets such as LAION-5B Although these datasets are typically compiled by nonprofit organizations, they may still include many copyrighted images.[40]

The process of using training data involves: gathering raw data from various sources (data collection); cleaning and standardizing the raw data for subsequent processing and analysis (data pre-processing); labeling the data to provide training data for machine learning (data annotation); extracting features from the annotated data (feature extraction); analyzing and learning from the training data (model training); producing the output (result generation).[41] The training datasets constructed through these methods involve the unauthorized use of copyrighted data and content, thus posing copyright infringement risks. Firstly, during the data collection phase, the use of training data may constitute an infringement of reproduction rights. Secondly, during the training process, GenAI often does not just temporarily copy data into the system but requires the data to be retained for a long period, which can involve the factual permanent reproduction of the works. Although there is currently debate among various countries about the legal nature of "temporary reproduction", there is a consensus that permanent reproduction should fall within the regulatory scope of reproduction rights.[42] Thirdly, in the data pre-processing phase, GenAI involves steps such as cleaning, standardizing, annotating, and feature extraction, which may infringe adaptation rights and other related rights. Finally, during the result generation phase, the use of training data may also infringe rights related to distribution. As GenAI automatically generates results based on the training data and presents them visually, the data must be transmitted over the internet, which, from a copyright law perspective, may also involve legal issues related to infringement.

Regarding the copyright issues related to the data content used in the training stage of GenAI, several key questions about the authorization and licensing of such data content need to be explored. If the copyright holder does not explicitly prohibit it, can GenAI developers assume implied consent to use copyrighted data content legally? Is it permissible for GenAI developers to automatically scrape internet data? How should conflicts between the use of training data and copyright be resolved? Answering these questions requires an in-depth discussion of the fair use doctrine. Fair use allows others to use copyrighted material without the copyright holder's consent or payment under specific conditions. At the national law level, U.S. law currently attempts to address the copyright

[28] Kingma and Welling (n 25).

[29] Cole Stryker and Mark Scapicchio, 'What Is Generative AI?' (*IBM*, 22 March 2024).

[30] R Bernardi, R Cakici and D Elliott, 'Automatic Description Generation from Images: A Survey of Models, Datasets, and Evaluation Measures' (2016) 55 Journal of Artificial Intelligence Research 409.

[31] Vrunda Gadesha and Eda Kavlakoglu, 'What Is Text Generation?' (*IBM.com*, 19 March 2024).

[32] Kristin J Dana, *Computational Texture and Patterns: From Textons to Deep Learning* (Springer 2022).

[33] Maryam M Najafabadi and others, 'Deep Learning Applications and Challenges in Big Data Analytics' (2015) 2 Journal of Big Data.

[34] Sophie J Nightingale and Hany Farid, 'AI-Synthesized Faces Are Indistinguishable from Real Faces and More Trustworthy' (2022) 119 Proceedings of the National Academy of Sciences.

[35] Gowthami Somepalli and others, 'Diffusion Art or Digital Forgery? Investigating Data Replication in Diffusion Models' [2022] arXiv:2212.03860.

[36] Nicholas Carlini and others, 'Extracting Training Data from Diffusion Models' [2023] arXiv:2301.13188.

[37] ibid.

[38] Tom Winter, 'Exploring Generative AI: Its Capabilities and Influence' (*SEOwind*, 20 May 2024).

[39] Somepalli and others (n 35).

[40] LAION, 'FAQ' (*laion.ai*).

[41] Yuliia Kniazieva, 'From Data to Dialogue: Data Annotation for Training AI Chatbots like ChatGPT' (*Labelyourdata.com*, 23 February 2023).

[42] Zohar Efroni, 'The Digital Reproduction Right' in Zohar Efroni (ed), *Access-Right: The Future of Digital Copyright Law* (Oxford University Press 2010).

protection issues of data content in the GenAI training stage through the application of case law. In contrast, EU law attempts to address this issue through provisions in the DSM Directive and the AI Act. At the international law level, WIPO conventions and other international law provide guiding principles on the fair use doctrine. These legislative and judicial attempts aim to strike a balance between the development of AI technology and the protection of copyright holders' interests.

3. The international and national legal framework for copyright protection related to GenAI data training

3.1. The international legal framework for copyright protection related to GenAI training

The data content for GenAI model training is sourced globally, encompassing nearly all data available on the internet, including but not limited to text, images, audio, and video. This means that copyright issues related to GenAI model training may be treated differently under the laws of various legal jurisdictions. The global scope of these copyright concerns underscores the need for international collaboration to develop appropriate solutions.

In recent years, amid the rapid development of GenAI, the UN has recognized the need to balance AI innovation with intellectual property protection. In December 2023, the UN Advisory Body on Artificial Intelligence released a report titled "Governing AI for Humanity", which, in its risk analysis, noted, "At the societal level, it encompasses the impact on […] creativity if intellectual property rights are not protected".[43] In March 2024, the UN General Assembly adopted a resolution titled "Seizing the Opportunities of Safe, Secure and Trustworthy Artificial Intelligence Systems for Sustainable Development", emphasizing the importance of "encouraging, where appropriate and relevant, the implementation of appropriate safeguards to respect intellectual property rights, including copyright-protected content, while promoting innovation".[44] Additionally, in July 2024, the UN General Assembly passed a resolution titled "Enhancing International Cooperation on Capacity-Building of Artificial Intelligence", stressing that "Member States should enjoy equal opportunities in the design, development, deployment, decommissioning, and use of artificial intelligence, while respecting intellectual property rights and promoting innovation".[45]

Additionally, the World Intellectual Property Organization (WIPO) has recognized the concerns surrounding AI and copyright. Since 2019, WIPO has been hosting the annual WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI). During the second session in July 2020, WIPO released the "Revised Issues Paper on Intellectual Property Policy and Artificial Intelligence", which explored issues related to AI training and copyright infringement, with Issue 8, titled "Infringement and Exceptions", specifically addressing these concerns.[46] For instance, the paper raises questions such as: "(iii) If the use of data subsisting in copyright works without authorization for machine learning is considered to constitute an infringement of copyright, should an explicit exception be made under copyright law or other relevant laws for the use of such data to train AI applications? […] (vi) Would any policy intervention be necessary to facilitate licensing if the unauthorized use of data subsisting in copyright works for machine learning is considered an infringement of copyright? Would the establishment of mandatory collective management societies facilitate this? Should remedies for infringement be limited to equitable remuneration?"[47] At the third session in November 2020, related issues, including "Infringement and Exceptions" (Issue 8), were discussed in greater depth.[48] The session records indicate ongoing differences of opinion among participants: "Speakers generally agreed that a balance is needed between the desire of AI developers to access large volumes of data and the rights of copyright owners whose protected works are included in that data. However, they were divided on possible solutions. Some believed that data mining articles or the fair use doctrine should provide adequate exceptions to copyright. Others warned against undermining the rights and interests of creators".[49] At the fourth session, copyright infringement issues related to GenAI model training remained a central topic, with arguments presented both for and against enforcing copyright protection on the data content used in GenAI training.[50] Concerns were raised about overly restrictive measures potentially stifling AI innovation, as well as the difficulties in tracking and penalizing infringement and fairly distributing benefits.[51] During the fifth session, many participants acknowledged that "there remain challenges that must be addressed internationally through collaboration, such as AI authorship or copyright in training data".[52]

In a WIPO factsheet titled "Generative AI: Navigating Intellectual Property", WIPO noted that "Some generative AI tools have been trained using materials scraped from the internet, including copyrighted works, personal information, biometric data, and harmful or illegal content. There is ongoing litigation over whether the scraping, downloading, and processing of these materials, the trained AI models, and their outputs constitute breaches of IP, privacy, and contract. Debates continue over how to balance the interests of IP owners and AI developers".[53] Furthermore, in the "Patent Landscape Report - Generative Artificial Intelligence", WIPO highlighted concerns about the growing use of GenAI models and tools, including issues such as copyright infringement, potential misuse, and the risk of job displacement.[54] On the topics of transparency and explanations, WIPO stressed the importance of documenting the training data used and the algorithms employed to avoid potential copyright infringement and to help users understand how GenAI models function.[55] Regarding the entertainment industry, WIPO noted that GenAI poses risks on complex legal, ethical, and technical fronts, including unresolved copyright issues related to GenAI training.[56]

From the above overview, it is evident that the UN and WIPO have increasingly focused on copyright issues related to GenAI model training in recent years. While the international community has yet to arrive at a definitive solution, there is a growing consensus on the importance of respecting intellectual property rights, including copyright-protected content, while promoting innovation.[57]

To date, no international agreement has specifically addressed this issue, and relevant discussions are still ongoing. Consequently, it is

[43] United Nations, 'Interim Report: Governing AI for Humanity' (2023).

[44] United Nations, 'Resolution Adopted by the General Assembly on 21 March 2024: 78/265. Seizing the Opportunities of Safe, Secure and Trustworthy Artificial Intelligence Systems for Sustainable Development' (2024).

[45] United Nations, 'Resolution Adopted by the General Assembly on 1 July 2024: 78/311. Enhancing International Cooperation on Capacity-Building of Artificial Intelligence' (2024).

[46] WIPO, 'WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI): Revised Issues Paper on Intellectual Property Policy and Artificial Intelligence' (2020).

[47] ibid.

[48] WIPO, 'WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI): Third Session' (2020).

[49] ibid.

[50] WIPO, 'WIPO Conversation on Intellectual Property (IP) and Frontier Technologies: Fourth Session' (2021).

[51] ibid.

[52] WIPO, 'WIPO Conversation on Intellectual Property (IP) and Frontier Technologies: Summary of the Fifth Session ' (2022).

[53] WIPO, 'Generative AI Navigating Intellectual Property' (2024).

[54] WIPO, 'Patent Landscape Report: Generative Artificial Intelligence' (2024).

[55] ibid.

[56] ibid.

[57] United Nations, 'Resolution Adopted by the General Assembly on 21 March 2024: 78/265. Seizing the Opportunities of Safe, Secure and Trustworthy Artificial Intelligence Systems for Sustainable Development' (n 44).

necessary to analyze the existing international copyright legal framework to lay the groundwork for exploring feasible solutions in the future. Several key international copyright agreements aim to protect the rights of creators of literary and artistic works. This article focuses on the limits on protection outlined in these international agreements to assess their applicability to GenAI model training. According to Sam Ricketson and Jane C. Ginsburg, the Berne Convention outlines three main approaches to limits on protection: subject matter limitations, restrictions on use, and uses subject to payment of compensation.[58] A typical example of subject matter limitations can be found in Article 2(8) of the Berne Convention. Some subject matter limitations are left to the discretion of national legislators, such as those in Article 2(4) and Article 2bis (1) of the Berne Convention. Regarding restrictions on use, key articles include Article 2bis (2), Article 9(2), Article 10, and Article 10bis of the Berne Convention. For uses subject to payment of compensation, typical examples can be found in Article 11bis (2) and Article 13 of the Berne Convention.

Article 9(2) of the Berne Convention states: "It shall be a matter for legislation in the countries of the Union to permit the reproduction of such works in certain special cases, provided that such reproduction does not conflict with a normal exploitation of the work and does not unreasonably prejudice the legitimate interests of the author". This article delegates the determination of limitations and exceptions to copyright protection to the domestic laws of member states, subject to the mandatory requirement of the well-known "three-step test". As a result, this article is also referred to as "2. Possible exceptions".[59] According to Professor Ginsburg, subsequent intellectual property treaties have largely preserved the approach to limitations and exceptions established by the Berne Convention.[60] Notable examples include Article 15 of the International Convention for the Protection of Performers, Producers of Phonograms, and Broadcasting Organizations (Rome Convention), Article 13 of the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS), Article 10 of the WIPO Copyright Treaty (WCT), Article 16 of the WIPO Performances and Phonograms Treaty (WPPT), and Article 13 of the Beijing Treaty on Audiovisual Performances (BTAP).

The above approach to limitations and exceptions grants countries significant legislative freedom, allowing them ample space to establish specific limitations and exceptions to copyright protection. However, this approach also introduces notable challenges, as it results in a lack of consensus on the rules governing copyright limitations and exceptions across different countries. Consequently, there are considerable differences in how these issues are addressed, leading to varying legal environments for GenAI model training and copyright protection in different countries.

The primary exception is the Marrakesh Treaty. Professor Senftleben believes that given the sensitive political climate surrounding the establishment of further international copyright law standards, the adoption of the Marrakesh Treaty is considered a miracle.[61] Despite numerous obstacles, the "Miracle of Marrakesh" shows that it is still possible to reach agreement on new international copyright rules.[62] Unlike the Berne Convention, the Marrakesh Treaty does not fully adopt the approach to limitations and exceptions established by the Berne Convention but instead imposes mandatory requirements in Article 4 ("National Law Limitations and Exceptions Regarding Accessible Format Copies"): "1.(a) Contracting Parties shall provide in their national copyright laws for a limitation or exception to the right of reproduction, the right of distribution, and the right of making available to the public as provided by the WIPO Copyright Treaty (WCT), to facilitate the availability of works in accessible format copies for beneficiary persons". Additionally, Article 11 of the Marrakesh Treaty explicitly requires that any limitations and exceptions arising from the implementation of the treaty must undergo strict scrutiny based on the "three-step test". This approach of combining mandatory requirements on limitations and exceptions with the three-step test has also been the subject of detailed academic discussion. Professor Senftleben argues that "The Marrakesh Treaty can lead to a nightmare scenario in which neither the goal of legal certainty nor the advantage of flexibility is realized. However, the Marrakesh Treaty also demonstrates how international norms could be structured to arrive at a sufficiently flexible and sustainable treaty framework for the regulation of L&Es at the national level".[63]

Whether the copyright issues related to GenAI model training can be addressed at the international level through collaborative efforts by the global community is an important topic worthy of discussion. Before exploring potential solutions within the framework of international copyright law, this article will first examine the approaches taken by specific legal jurisdictions to highlight the different legislative approaches to GenAI model training. In specific legal jurisdictions, two distinct legislative approaches have emerged: the industry-oriented path, exemplified by U.S. law, which tends to promote industrial development, and the rights-oriented path, represented by EU law, which focuses on protecting fundamental rights. The following two parts will provide a detailed analysis of these approaches.

3.2. U.S. industry-oriented copyright law and the fair use doctrine

In U.S. copyright law, the doctrine of fair use is a crucial exception and limitation mechanism designed to balance copyright protection with public interest. This doctrine allows third parties to legally use copyrighted works under specific conditions without constituting infringement. The concept of fair use originated from case law and was formally established in Section 107 of the 1976 Copyright Act, becoming the central standard for determining whether a particular use of content by GenAI constitutes infringement.

3.2.1. Traditional factors for determining fair use

The determination of fair use requires a comprehensive consideration of four factors. Firstly, the purpose and character of the use should be considered. Although Section 107 of the U.S. Copyright Act does not specify the relative importance of these four factors, allowing judges to apply them at their discretion based on individual cases, judicial practice has shown that the first factor has become the primary consideration in determining fair use in the U.S. Since *Campbell v. Acuff-Rose Music, Inc.*,[64] U.S. courts have considered "transformative use" as the standard for judging fair use, particularly when the commercial use significantly enhances social benefits, regardless of its commercial nature. The concept of "transformative use" involves some degree of adaptation, transformation, or conversion of the original work to create new expression, meaning, or value.[65] As anecdotal evidence of the US

[58] Sam Ricketson and Jane C Ginsburg, 'Restrictions on the Exercise of Rights: Limitations and Exceptions (Articles 2bis (2), 9(2), 10, 10bis (2), 13, 17; Later Agreements)' in Sam Ricketson and Jane C Ginsburg (eds), *International Copyright and Neighbouring Rights: The Berne Convention and Beyond* (Oxford University Press 2022).

[59] Berne Convention for the Protection of Literary and Artistic Works, September 9, 1886, as amended on September 28, 1979.

[60] Ricketson and Ginsburg (n 58).

[61] Martin Senftleben, 'A Copyright Limitations Treaty Based on the Marrakesh Model: Nightmare or Dream Come True?' in Shyamkrishna Balganesh, Ng-Loy Wee Loon and Haochen Sun (eds), *The Cambridge Handbook of Copyright Limitations and Exceptions* (Cambridge University Press 2021).

[62] ibid.

[63] ibid.

[64] *Campbell v Acuff-Rose Music, Inc,* 510 US 569 (1994).

[65] Pierre N Leval, 'Toward a Fair Use Standard' (1990) 103 Harvard Law Review 1105.

approach being industry-oriented, courts have increasingly ruled that even minimal changes can be considered transformative.[66] This trend has led to calls for a reconsideration of the role that transformative use plays in fair use analysis.[67] Consequently, defining the precise boundaries of what constitutes transformative use is crucial to the overall analysis but remains challenging to determine concretely.[68] Secondly, the nature of the copyrighted work should be considered.[69] This factor examines whether the used work is a factual narrative or a highly creative fictional work. Generally, the more creative a work is, the more it deserves legal protection. The third factor is the amount and substantiality of the portion used in relation to the copyrighted work as a whole.[70] This factor requires both quantitative and qualitative analysis of the usage. Even if the amount used is relatively small, using the "heart" of the work may still not be considered fair use. Conversely, if a larger portion is used but it serves only an auxiliary purpose and does not constitute the core content of the work, it may still qualify as fair use. The last factor is the effect of the use upon the potential market for or value of the copyrighted work, which assesses whether the use adversely affects the normal market for the copyrighted work or its potential market.[71] If the new work does not significantly replace the market demand for the original work, it is more likely to be deemed fair use. However, if the use reduces the copyright holder's revenue, it may be considered unreasonable use.

3.2.2. Non-expressive fair use

Professor Benjamin I. W. Sobel applied the concept of "non-expressive fair use", building on the four traditional factors of fair use, to determine whether large-scale, unauthorized copying by computers constitutes fair use.[72] He posited that the concept of non-expressive fair use has two core premises. First, traditional computerized analysis interacts with the non-expressive parts of human works, meaning that traditional machines cannot infringe on copyrighted expressions. Therefore, as long as the machine's acquisition of the work is not intended to facilitate human engagement with the work's expression, such acquisition has a non-expressive purpose. Second, since copyright holders' rights do not cover the non-expressive parts of their works, traditional machines' use of these non-expressive parts does not impact the potential market for the work in a way that substantively affects copyright law. However, GenAI using deep learning may not meet these two premises of non-expressive fair use. Professor Matthew Sag believes that any more than a trivial amount of memorization of significant details from works in the training data during the GenAI data training stage jeopardizes any fair use claim these models have under a theory of non-expressive fair use.[73] GenAI can extract value not only from the factual aspects of a work but also from its expressive aspects. It might also replace human authors and their markets with cheaper and more efficient automated technology, thereby impacting the realization of copyright law's legislative purposes.[74]

Although there has not been a judicial ruling on the copyright issues related to GenAI data training, several U.S. court cases involving algorithmic technologies—such as reverse engineering, search engines, plagiarism detection software, and book digitization—have applied the concept of non-expressive fair use to determine copyright exceptions, favoring the technology industry. In *Ninth Circuit's Sega v. Accolade decision*,[75] Accolade sought to create games compatible with Genesis but did not wish to enter a licensing agreement with Sega. To solve technical human-machine interaction details, Accolade copied and analyzed three Sega game software programs, performing reverse engineering to replicate the functional computer code that allowed game cartridges to be compatible with the console. Accolade's reverse engineering process inevitably produced several verbatim copies of Sega's game source code but ultimately included only the functional code related to Genesis compatibility. The Ninth Circuit Court of Appeals ruled that Accolade's intermediate copying through reverse engineering was necessary to access the non-expressive elements of the work—Genesis compatibility functions—and the final product was a work created in Accolade's own expression.[76] Similarly, in *Sony v. Connectix*, the court emphasized that copyright law protects expression, not the ideas or functions of a software program. Thus, the fair use doctrine protects the public's access to the ideas and functional content embedded in computer software programs.[77]

Moreover, *Kelly v. Arriba*[78] explored the rules for determining non-expressive fair use in the context of search engines. Arriba Soft Corp. operated a search engine that generated thumbnails collected via crawler technology based on search queries. Arriba had copied the expressive content of Kelly's works and displayed some of this expressive content to the public. However, the court held that Arriba's low-resolution thumbnails were not artistic expressions and did not inhibit artistic creativity because they were not used for illustrative or artistic purposes and thus did not replace the demand for the original works. In the context of plagiarism detection services, the court in *A. V. ex rel. Vanderhye v. iParadigms* held that if a plagiarism report displays source documents as needed, this optional feature constitutes expressive use but is protected under fair use because its purpose is to provide evidence for fraud litigation.[79]

Furthermore, the case of *Authors Guild v. Google Inc*. analyzed the application of non-expressive fair use in the context of book digitization.[80] Google scanned over twenty million books, creating a machine-readable text corpus that supports its commercial book search engine, Google Books. Authorized books were displayed in a "snippet view" format, showing a limited number of full-text pages. The court ruled that although the "snippet view" displayed text around users' search terms, it advanced Google's transformative purpose by providing context for the terms without revealing enough content to threaten the authors' copyright interests. Google Books could potentially harm authors' markets, but this harm generally occurs in situations where the interest is not protected by copyright.

3.2.3. An extreme example of the U.S. industry-oriented approach: japanese copyright law

In addressing copyright issues in GenAI model training, Japan has also adopted an industry-oriented approach to copyright. Article 30–4 of the Japanese Copyright Act, amended in 2018, provides an exception for the use of copyrighted works in GenAI model training. Article 30–4 states: "It is permissible to exploit a work in any manner and to the extent considered necessary in the following cases, or in any other case where the intent is not for a person to personally enjoy or cause another

[66] ibid; Clark D Asay, Arielle Sloan and Dean Sobczak, 'Is Transformative Use Eating the World?' (2020) 61 Boston College Law Review 905.
[67] David Nimmer, *Nimmer on Copyright* (Matthew Bender Elite Products 2023).
[68] Asay, Sloan and Sobczak (n 66).
[69] Leval (n 65).
[70] ibid.
[71] ibid.
[72] Sobel (n 21).
[73] Sag (n 17).
[74] Matthew Sag, 'The New Legal Landscape for Text Mining and Machine Learning' (2019) 66 Journal of the Copyright Society of the U.S.A. 291.
[75] *Sega Enters. v. Accolade, Inc.,* 977 F.2d 1510, 1514 (9th Cir. 1992). (n 12).
[76] James Grimmelmann, 'Copyright for Literate Robots' (2015) 101 Iowa Law Review 657.
[77] *Sony Computer Ent. v. Connectix Corp.,* 2203 F.3d 596, 608 (9th Cir. 2000). (n 12).
[78] *Kelly v. Arriba Soft Corp.,* 336 F.3d 811 (9th Cir. 2003). (n 12).
[79] *A.V. ex rel. Vanderhye v. iParadigms, LLC,* 562 F.3d 630, 644-45 (4th Cir. 2009) (n 12).
[80] *Authors Guild, Inc v Google Inc (SDNY2013)*; 'Authors Guild, Inc. v. Google Inc., 804 F.3d 202, 2nd Cir. (N.Y.2015).'

person to enjoy the thoughts or sentiments expressed in the work. However, this exception does not apply if the action would unreasonably prejudice the interests of the copyright owner, considering the nature or purpose of the work or the circumstances of its use: […] (ii) if it is done for data analysis (meaning the extraction, comparison, classification, or other statistical analysis of the constituent language, sounds, images, or other elemental data from a large number of works or a large volume of other such data; […])".

According to the official interpretation by the Agency for Cultural Affairs of Japan, "Based on these provisions, it would become possible to widely carry out acts without the purpose of enjoying the thoughts or sentiments expressed in a work, such as the act of recording a work in a database as training data for the development of AI, without the authorization of the right holder".[81] Kung-Chung Liu and Shufeng Zheng analyze that, "By using broadly termed words like 'exploit' and 'in any way', this article allows a wide range of uses in data mining activities. Some have also expressed the opinion that this article permits the business model of creating a learning data set consisting of third parties' works, and then transferring and disclosing it to an unspecified third party who intends to develop AI".[82] Professor Tatsuhiro Ueno also believes that Article 30–4(ii) allows any exploitation of works for text and data mining.[83]

Article 30–4(ii) of the Japanese Copyright Act, which grants a broad exception for the use of copyrighted works in GenAI model training, has raised concerns among creators in Japan's creative industries about their works being used by GenAI model providers without authorization or compensation.[84] For instance, Japanese illustrator Momoji Mokume has pessimistically remarked that Japan is turning into a paradise for piracy and machine learning, expressing fears that illustrators' jobs may vanish and that society could lose its culture of innovation.[85]

3.3. EU rights-oriented copyright law

In contrast to the industry-oriented approach of U.S. copyright law, EU copyright law tends to be rights-oriented, favoring the interests of copyright holders. Developing and training GenAI models require access to vast amounts of text, images, videos, and other data content, which may be protected by copyright and/or sui generis database rights. Under EU copyright law, generally, using copyrighted content as training data for commercial purposes requires authorization from the relevant rightholders unless an exception or limitation applies. Furthermore, EU copyright law does not tolerate the lack of transparency from many GenAI model providers, who do not disclose the data content used to train their models, leaving rightholders unable to assert their rights. EU rights-oriented copyright law will be analyzed through the lens of the DSM Directive and the AI Act, both of which are applicable to GenAI.

3.3.1. The DSM directive

The DSM Directive introduces exceptions to the rights for reproducing and extracting lawfully accessible works and other subject matter for the purposes of text and data mining, which are particularly relevant to GenAI data training.[86] "Text and data mining" is defined by Article 2(2) of the DSM Directive as "any automated analytical technique aimed at analysing text and data in digital form in order to generate information which includes but is not limited to patterns, trends and correlations". This is a broad definition that may encompass most activities reliant on automated analytical techniques for processing information and extracting meaning, including the machine learning algorithms used by GenAI.[87] This interpretation is further supported by Recital 18, which states: "In addition to their significance in the context of scientific research, text and data mining techniques are widely used by both private and public entities to analyze large amounts of data across different areas and for various purposes, including the development of new applications or technologies". Moreover, the two text and data mining exceptions under the DSM Directive are established based on their intended purpose. Article 3 regulates the text and data mining exception for scientific research, while Article 4 provides for an exception or limitation for commercial text and data mining. These two exceptions only apply to reproductions and extractions of lawfully accessible works and other subject matter for the purposes of text and data mining. The rights of distribution and communication are not included within these exceptions.[88] This means that GenAI model providers can extract content from databases or copy other protected materials, but they are not permitted to publicly share these contents.

Regarding Article 3, Article 3(1) of the DSM Directive stipulates that member states should provide an exception for research organizations and cultural heritage institutions to carry out reproductions and extractions for text and data mining of works or other subject matter to which they have lawful access for scientific research purposes. Under Article 3(3), member states should allow copyright holders to implement measures to ensure the security and integrity of their networks and databases containing their works or other subject matter, provided that these measures do not exceed what is necessary to achieve this goal.[89]

According to Article 4, rightholders can choose to reserve their rights and prevent the use of their content in GenAI data training, unless it is for scientific research purposes. If rightholders explicitly retain these rights in an appropriate manner, providers of GenAI models must obtain authorization from the rightholders to perform GenAI data training on these works.[90]

In specific, Article 4(1) stipulates that member states shall provide an exception or limitation for reproductions and extractions of works or other subject matter to which they have lawful access for the purposes of text and data mining. This means that GenAI model providers must have lawful access to the works and other subject matter used to train their models. While Recital 14 applies specifically to research organizations and cultural heritage institutions, its interpretation of "lawful access" is

[81] 'Agency for Cultural Affairs of Japan', 'Regarding the Act Partially Amending the Copyright Act (Act No. 30 of 2018)' (*Agency for Cultural Affairs of Japan*, 2018) <bunka.go.jp/english/policy/copyright/amendments_2018> accessed 12 August 2024.

[82] Kung-Chung Liu and Shufeng Zheng, 'Protection of and Access to Relevant Data - General Issues' in Jyh-An Lee, Reto Hilty and Kung-Chung Liu (eds), *Artificial Intelligence and Intellectual Property* (Oxford University Press 2021).

[83] Tatsuhiro Ueno, 'The Flexible Copyright Exception for "Non-Enjoyment" Purposes - Recent Amendment in Japan and Its Implication ' (2021) 70 GRUR International 145.

[84] Kana Inagaki and David Keohane, 'Japan's Copyright Rules Draw AI Groups - and Alarm from Creators' *Financial Times* (22 July 2024).

[85] ibid.

[86] Katharina de la Durantaye, 'Garbage In, Garbage Out. Regulating Generative AI Through Copyright Law' [2023] ZUM 645; Christiane Stuetzle, 'AI Trends for 2024 - EU Copyright Law: AI Act on the Home Stretch' (*MOFO TECH*, 16 January 2024).

[87] Martin Senftleben, 'Copyright Data Improvement for AI Licensing - The Role of Content Moderation and Text and Data Mining Rules' (2024) <ssrn.com/abstract=4817796> accessed 31 July 2024.

[88] ibid.

[89] Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on Copyright and Related Rights in the Digital Single Market and Amending Directives 96/9/EC and 2001/29/EC.

[90] Eleonora Rosati, 'Copyright as an Obstacle or an Enabler? A European Perspective on Text and Data Mining and Its Role in the Development of AI Creativity' (2020) 27 Asia Pacific Law Review 198.

generally considered applicable to GenAI as well.[91] According to Recital 14, "lawful access" should be understood as including access to "content based on an open access policy or through contractual arrangements", such as subscriptions or open access licenses, as well as to "content that is freely available online". Therefore, the DSM Directive does not opt for a general exemption for text and data mining based on the distinction between the use of "works as works" and "works as data", which serves as a metaphor for the belief that text and data mining fall outside the traditional concept of reproduction.[92] As Rosanna Ducato and Alain Strowel argue, "when acts of reproduction are carried out for the purpose of search and TDM, the work, although it might be reproduced in part, is not used as a work: the work only serves as a tool or data for deriving other relevant information".[93] On the contrary, Article 4(1) of the DSM Directive places text and data mining under the umbrella of the right of reproduction enjoyed by rightholders.[94]

Article 4(2) states that reproductions and extractions made in accordance with paragraph 1 may be retained for as long as is necessary for text and data mining. This suggests that data content used for training GenAI models should be deleted once the training phase is completed. It also implies that appropriate security measures must be implemented to ensure the secure retention of these copies.[95]

Article 4(3) specifies that the exception or limitation provided for in paragraph 1 shall apply on condition that the use of works and other subject matter has not been expressly reserved by their rightholders in an appropriate manner, such as machine-readable means in the case of content made publicly available online. This means that rightholders have the option to expressly reserve their rights and opt out of allowing their works to be used for text and data mining purposes. After an express reservation of rights, GenAI model providers will no longer be able to use these works for GenAI model training, updates, or the creation of new versions.[96] Conversely, all copyrighted works that have not been actively managed by the rightholders and lawfully made publicly available may be freely mined and used for training GenAI models. The concept of "content made publicly available online" generally refers to any content accessible to the public, such as digital music files available on both free and paid music streaming services. For content that has been made publicly available online, according to Recital 18, it should only be considered appropriate to reserve those rights by the use of machine-readable means, including metadata and terms and conditions of a website or a service. Metadata includes files such as robots.txt file, ai.txt file, or their equivalents.[97] A concise and easily accessible summary of the terms and conditions of a website or service, provided in a machine-readable format, may enable machines to read and interpret statements reserving text and data mining rights to the rightholders. To emphasize, the language of the opt-out statement must be clear and unambiguous, providing a detailed description of the uses reserved by the rightholders. Other uses should not be affected by the reservation of rights for the purposes of text and data mining. Moreover, in the case of content that has not been made publicly available online, it can be appropriate for rightholders to reserve their rights by other means, such as contractual agreements or a unilateral declaration (e.g., a press release). Examples of content not publicly available online include digital books in private libraries not connected to the internet or CDs containing digital audio recordings. Finally, according to Recital 18, "[r]ightholders should be able to apply measures to ensure that their reservations in this regard are respected". This may imply that, when exercising the right to opt-out, rightholders can implement technical protection measures, such as digital rights management solutions, to control the use and distribution of their works. As Professor Martin Senftleben stated, the rights reservation option in Article 4(3) could lead to the development of machine-readable rights reservation protocols that represent the various positions of different rightsholders.[98] Despite efforts to develop text and data mining reservation protocols,[99] there is currently no widely recognized machine-readable opt-out reservation protocol, nor a standardized method for expressing rights reservations for content not publicly available online. This issue makes it difficult for rightholders to express their rights reservations consistently across the Internet.

Article 4(4) clarifies that Article 4 shall not affect the application of Article 3.[100] This means that the rightholders' option to opt-out under Article 4(3) applies exclusively to text and data mining for purposes other than scientific research, primarily targeting commercial text and data mining.

Although there are no GenAI-related judicial cases under Article 4 of the DSM Directive, Eleonora Rosati provides a useful example to illustrate the practical application of this Article.[101] In 2017, SONY CSL Research Lab released the song "Daddy's Car", entirely composed by FlowMachines. This song was created in the style of The Beatles, which required data mining The Beatles' repertoire to train FlowMachines and produce a song in that style. From the perspective of Article 4, such a project would only be permissible if there was lawful access to the repertoire, the relevant rightholders had not reserved the right to make reproductions for text and data mining purposes, and any copies made were retained only for the duration necessary for text and data mining. Failing to meet these conditions, such an unauthorized project could constitute infringement.

As another important function, Article 4 aims to protect fair competition in the international market for GenAI.[102] Some argue that allowing rightholders to reserve their rights could potentially drive GenAI model providers to seek other countries with more lenient copyright exceptions for data training. Consequently, EU copyright holders might not receive the expected compensation, leading to a lose-lose situation for both EU rightholders and the EU GenAI industry.[103] In fact, under Article 4, any provider introducing GenAI models to the EU market must comply with this obligation, regardless of the jurisdiction where the copyright-related actions during the GenAI model's training occurred. This is crucial for ensuring a fair competitive environment among GenAI model providers. No provider should gain a competitive advantage in the EU market by applying copyright standards that are lower than those required by the EU. Therefore, providers of GenAI models should develop policies to comply with EU laws on copyright and related rights, particularly identifying and respecting the rightholders' reservations expressed under Article 4(3).

[91] Joao Pedro Quintais, 'The New Copyright in the Digital Single Market Directive: A Critical Look' (2020) 1 European Intellectual Property Review.

[92] Martin Senftleben, 'Compliance of National TDM Rules with International Copyright Law - An Overrated Nonissue?' (2022) 53 IIC - International Review of Intellectual Property and Competition Law 1477.

[93] R Ducato and A Strowel, 'Ensuring Text and Data Mining: Remaining Issues with the EU Copyright Exceptions and Possible Ways Out' (2021) 43 European Intellectual Property Review 322.

[94] P Keller, 'Protecting Creatives or Impeding Progress? Machine Learning and the EU Copyright Framework' (*Kluwer Copyright Blog*, 20 February 2023).

[95] Quintais (n 91).

[96] Peter Mezei, 'A Saviour or a Dead End? Reservation of Rights in the Age of Generative AI' (2024).

[97] Martin Senftleben, 'Generative AI and Author Remuneration' (2023) 54 IIC - International Review of Intellectual Property and Competition Law 1535.

[98] Senftleben, 'Copyright Data Improvement for AI Licensing - The Role of Content Moderation and Text and Data Mining Rules' (n 87).

[99] W3C, 'TDM Reservation Protocol (TDMRep)' (10 May 2024).

[100] Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on Copyright and Related Rights in the Digital Single Market and Amending Directives 96/9/EC and 2001/29/EC.

[101] Rosati (n 90).

[102] Quintais (n 91).

[103] Senftleben, 'Generative AI and Author Remuneration' (n 97).

3.3.2. The AI Act

In addition to the DSM Directive, EU rights-oriented copyright law can also be analyzed through the lens of the AI Act, which applies to GenAI. As Christophe Geiger and Vincenzo Iaia clarify, the AI Act initially did not address copyright issues.[104] However, tensions between GenAI model providers and copyright holders prompted the European Parliament to include some limited considerations regarding the copyright aspects of GenAI.[105] The GenAI models discussed in this article fall under the concept of "general-purpose AI models" as defined in the AI Act. According to Recital 97 of the AI Act, general-purpose AI models are typically trained on large amounts of data using various methods, such as self-supervised, unsupervised, or reinforcement learning. These models can be further modified or fine-tuned into new models. The concept of "general-purpose AI models" differs from that of AI systems.[106] While general-purpose AI models are fundamental components of AI systems, they do not constitute AI systems themselves. Additional components, such as user interfaces, are required for general-purpose AI models to become fully functional AI systems. Typically, general-purpose AI models are integrated into AI systems, becoming a part of them. For instance, multimodal large language models are a type of general-purpose AI model that can be integrated into AI systems, such as chatbots or text-to-speech applications. The transparency rules applicable to general-purpose AI systems under the AI Act should also apply when these models are integrated into or become part of AI systems.

Regarding copyright-related aspects, the AI Act primarily requires transparency when GenAI uses copyrighted content for model training.[107] The transparency obligations for GenAI model providers are mainly detailed in Article 53(1) of the AI Act. GenAI model providers should put in place a policy to comply with EU law on copyright and related rights, and in particular to identify and comply with a reservation of rights expressed pursuant to Article 4(3) of the DSM Directive.[108] Additionally, they must draw up and make publicly available a sufficiently detailed summary about the content used for training the GenAI models, according to a template provided by the AI Office.[109] Apart from these two transparency obligations, GenAI model providers shall draw up and keep up-to-date the technical documentation of the model,[110] and must also draw up, keep up-to-date and make available information and documentation to providers of AI systems who intend to integrate the GenAI model into their AI systems.[111]

Under the AI Act, transparency obligations for GenAI model providers apply once these models are placed on the market. According to Article 3(9), "placing on the market" refers to the first instance a model is made available in the EU market. General-purpose AI models (including GenAI models) can be placed on the market in various ways, including through libraries, application programming interfaces (APIs), direct downloads, or physical copies.[112] When a provider integrates their general-purpose AI model into their own AI system that is then made available on the market or put into service, the model should be considered placed on the market.[113] Consequently, the transparency obligations outlined in the AI Act for models should continue to apply in addition to those for AI systems. However, these transparency obligations do not apply when a model is used solely for internal processes that are not essential for providing a product or service to third parties and do not affect the rights of natural persons.[114] GenAI models that are used exclusively for research, development, and prototyping activities before being placed on the market are excluded. Nonetheless, this does not exempt the obligation to comply with the AI Act when a model is placed on the market following such activities. Importantly, given their potentially significant negative effects, general-purpose AI models with systemic risk should always be subject to the relevant obligations under this AI Act.[115] The following analysis will examine the transparency obligations and related issues.

First, according to Article 53(1)(c), providers of general-purpose AI models are required to "put in place a policy to comply with Union law on copyright and related rights, and in particular to identify and comply with, including through state-of-the-art technologies, a reservation of rights expressed pursuant to Article 4(3) of Directive (EU) 2019/790″. This provision, along with Recital 105 of the AI Act, connects the use of copyrighted works for training GenAI models with exceptions or limitations for commercial text and data mining, thereby settling the debate on whether Article 4 of the DSM Directive applies to GenAI. As Christophe Geiger and Vincenzo Iaia argue, the European legislator was aware that training GenAI models could involve works protected by copyright or sui generis database rights, potentially infringing on the exclusive reproduction right.[116] According to Recital 105, general-purpose AI models, particularly large GenAI models capable of generating text, images, and other content, present unique innovation opportunities but also challenges to artists, authors, and other creators regarding how their creative content is created, distributed, used, and consumed. The development and training of such models require access to vast amounts of text, images, videos, and other data. Text and data mining techniques may be extensively used in this context for retrieving and analyzing such content, which may be protected by copyright and related rights. Any use of copyright-protected content requires authorization from the concerned rightsholder unless relevant copyright exceptions and limitations apply.[117] The DSM Directive introduced exceptions and limitations allowing reproductions and extractions of works or other subject matter for text and data mining under certain conditions. Under these rules, rightsholders may choose to reserve their rights over their works or other subject matter to prevent text and data mining unless done for scientific research purposes. Where the right to opt out has been expressly reserved in an appropriate manner, authorization from rightsholders is required for GenAI models to conduct text and data mining on such works.[118] As Christophe Geiger and Vincenzo Iaia suggest, the transparency obligations outlined in Article 53 of the AI Act were likely introduced to enable rightsholders to more effectively exercise the opt-out right from the text and data mining exception established by Article 4(3) of the DSM Directive and to ensure its enforcement.[119] Without a reporting obligation, authors and rightsholders have no way to trace the supply chain behind outputs generated by GenAI models.

Secondly, Article 53(1)(d) of the AI Act requires GenAI model providers to prepare and publicly release a sufficiently detailed summary of the content used for training their models, following a template provided by the AI Office. Regarding the necessity for such a detailed summary, Recital 107 of the AI Act clarifies: "In order to increase transparency on the data that is used in the pre-training and training of general-purpose AI models, including text and data protected by copyright law, it is adequate that providers of such models draw up and make publicly available a sufficiently detailed summary of the content used

[104] Christophe Geiger and Vincenzo Laia, 'The Forgotten Creator: Towards a Statutory Remuneration Right for Machine Learning of Generative AI' (2024) 52 Computer Law & Security Review.
[105] ibid.
[106] Recital 97, the AI Act.
[107] Alexander Peukert, 'Copyright in the Artificial Intelligence Act - A Primer' (2024) 73 GRUR International 497.
[108] Article 53(1)(c), the AI Act.
[109] Article 53(1)(d), the AI Act.
[110] Article 53(1)(a), the AI Act .
[111] Article 53(1)(b), the AI Act.
[112] Recital 97, the AI Act.
[113] ibid.
[114] ibid.
[115] ibid.
[116] Geiger and Laia (n 104).
[117] Recital 105, the AI Act.
[118] ibid.
[119] Geiger and Laia (n 104).

for training the general-purpose AI model". The transparency requirement in Article 53(1)(d) of the AI Act enhances the practical applicability of Article 4 of the DSM Directive, which governs the exceptions or limitations related to text and data mining. By mandating that providers of general-purpose AI models supply a sufficiently detailed summary, rightsholders can determine whether their content has been used as training material and timely assert their reservation rights to GenAI model providers.[120]

In light of the fact that the training of GenAI models may involve large amounts of data scraped from the internet, identifying all the data used can be difficult or even impossible. Therefore, this "sufficiently detailed summary" should not attempt to list every work used for training the models individually. Generally, it is not recommended to adopt the approach proposed in the recent draft bill introduced in France on September 12, 2023, which requires listing the names of all authors whose works were used in the training of GenAI models.[121] Such an approach could be highly detrimental to the development of the innovative GenAI sector in the EU. Instead, while considering the need to protect trade secrets and confidential business information, this summary should aim to be generally comprehensive in scope rather than technically detailed.[122] This approach would facilitate the ability of parties with legitimate interests, including copyright holders, to exercise and enforce their rights under EU law.[123] For example, the summary could list the main data collections or sets used in training the model, such as large private or public databases or data archives, and provide a narrative explanation of other data sources employed.[124] However, in certain cases, merely listing the main data collections or sets may not suffice. For instance, a GenAI model provider might select a set of modern songs to fine-tune the model for creating contemporary music. In such scenarios, the summary may need to indicate which authors' works were used, though it need not specify each work individually. An overly general reference to databases, collections, or libraries might not be sufficient for rights holders to determine whether the model's training complies with Article 4 of the DSM Directive, thereby enabling them to exercise and protect their rights. Some scholars argue that meeting the transparency obligations regarding the works used appears quite unfeasible due to the low, still inconsistent threshold of originality, the fragmentation of copyright across various jurisdictions and multiple ownerships, the absence of a mandatory registration process, and the generally inadequate state of ownership metadata.[125]

GenAI models are periodically modified or fine-tuned. According to Recital 97 of the AI Act, these models may undergo further modifications or fine-tuning, resulting in new models. In other words, substantial modification or fine-tuning can lead to the creation of a new model, which will then be subject to the transparency requirements of the AI Act. This serves as an important anti-circumvention clause. If a GenAI model is provided with new training data for substantial modification or fine-tuning, the transparency obligations will be reinstated, and GenAI model providers will be required to disclose a summary of the new training data.

Regarding the transparency obligations of GenAI model providers to establish a policy that complies with EU copyright law and to publicly disclose a summary of the content used for training, the AI Office should monitor whether these obligations are being met.[126] However, the AI Office should not be responsible for verifying or conducting a work-by-work assessment of the training data for copyright compliance.[127] The AI Act does not alter the enforcement of copyright rules as provided under EU law, which will continue to be a matter of private law. Furthermore, it is appropriate for the AI Office to provide a template for the summary. This template should be simple, effective, and allow providers to present the required information in narrative form.[128] According to a report by Reuters, the AI Office plans to release a template in early 2025, following consultations with stakeholders.[129] Some scholars have suggested that the template could include the following elements: general information (such as size and ethical review), data sources and data sets (including information on data collection, date range, data cutoffs, opt-outs, the legal basis for data collection and processing, anonymization techniques, intermediaries, and a list of data sets), data diversity, and data processing.[130]

Thirdly, Article 53(1)(a) and Article 53(1)(b), along with Annexes XI and XII of the AI Act, establish the transparency obligations of GenAI model providers concerning documentation.[131] Under Article 53(1)(a), GenAI model providers shall draw up and keep up-to-date technical documentation of the model, which shall contain, at a minimum, the information set out in Annex XI for the purpose of providing it, upon request, to the AI Office and the national competent authorities. Annex XI outlines the minimum content required, including information about the types and sources of data and the methods used for obtaining and selecting the data. This provision is particularly useful in practice, as it enables the AI Office and national regulatory authorities to gain insights into the training data and processes, thereby countering any claims that providing such information is impossible.

Article 53(1)(b) of the AI Act requires GenAI model providers to draw up, keep up-to-date and make available information and documentation for AI system providers who intend to integrate the GenAI model into their systems. This information and documentation, while respecting and protecting intellectual property rights, confidential business information, and trade secrets in accordance with EU and national law, must enable AI system providers to clearly understand the capabilities and limitations of the GenAI model and ensure compliance with their obligations under the AI Act. Additionally, the information and documentation must include, at a minimum, the elements specified in Annex XII of the AI Act. According to Annex XII, the information must contain at least a general description of the GenAI model and an overview of its elements and development process. Specifically, the general description should include: the tasks the model is designed to perform and the type and nature of AI systems it can be integrated into; the applicable acceptable use policies; the release date and distribution methods; how the model interacts with hardware or software not part of the model itself; the versions of relevant software related to the use of the GenAI model; the architecture and number of parameters; the modality and format of inputs and outputs; the model's licensing information. Additionally, the description of the model's elements and development process should include: the technical means required for integrating the GenAI model into AI systems; the modality and format of the inputs and outputs, along with their maximum size; information on the data used for training, testing, and validation, including the type and provenance of data and curation methodologies.

GenAI model providers can avoid the transparency obligations outlined in Article 53(1)(a) and Article 53(1)(b) of the AI Act by offering a

[120] Peukert (n 107).
[121] Geiger and Laia (n 104).
[122] Recital 107, the AI Act.
[123] ibid.
[124] ibid.
[125] Joao Pedro Quintais, 'Generative AI, Copyright and the AI Act' (*Kluwer Copyright Blog*, 9 May 2023).
[126] Recital 108, the AI Act.
[127] ibid.
[128] Recital 107, the AI Act.
[129] Martin Coulter, 'EU's New AI Rules Ignite Battle over Data Transparency' *Reuters* (13 June 2024).
[130] Zuzanna Warso, Maximilian Gahntz and Paul Keller, 'Sufficiently Detailed? A Proposal for Implementing the AI Act's Training Data Transparency Requirement for GPAI' (*OPENFUTURE*, June 2024).
[131] Gustavo Ghidini, 'IP and AI - For a Balanced, Non-Protectionist Stance' [2024] GRUR International.

GenAI model under a free and open-source license that permits access, usage, modification, and distribution of the model. Additionally, they must publicly disclose the model's parameters, including the weights, information on the model architecture, and details on model usage.[132] However, this exception does not apply to general-purpose AI models that pose systemic risks.

However, as Recital 104 of the AI Act clarifies, in any case, given that the release of GenAI models under free and open-source licence does not necessarily reveal substantial information on the data set used for the training or fine-tuning of the model and on how compliance of copyright law was thereby ensured, the exception provided for GenAI models from compliance with the transparency-related requirements should not concern the obligation to produce a summary about the content used for model training and the obligation to put in place a policy to comply with EU copyright law, in particular to identify and comply with the reservation of rights pursuant to Article 4(3) of the DSM Directive.

Finally, in other relevant aspects, the following points should be noted. As mentioned above, the AI Act classifies GenAI models into three categories: *standard* GenAI models, GenAI models released under a free and open-source license, and GenAI models with systemic risks. Some scholars argue that this three-tier classification system may not adequately account for the nuances of downstream applications, potentially resulting in overly broad or narrow risk categories.[133]

It should be noted that, according to Article 53(4) of the AI Act, GenAI model providers may rely on codes of practice as defined in Article 56 to demonstrate compliance with the transparency obligations outlined in Article 53(1). The AI Office can invite GenAI model providers to develop codes of practice that specify "the adequate level of detail for the summary about the content used for training".[134] The AI Office may also invite all GenAI model providers to participate in the development of these codes of practice. This reliance on codes of practice should continue until a harmonized standard is published. Compliance with European harmonized standards grants providers a presumption of conformity, insofar as those standards cover the relevant obligations. GenAI model providers who do not adhere to an approved code of practice or comply with a European harmonized standard must demonstrate alternative adequate means of compliance for assessment by the Commission.

Regarding the extraterritorial application of transparency obligations for GenAI model providers in the AI Act, to ensure a level playing field in the EU market, if a GenAI model is developed outside the EU and then integrated into applications provided by EU providers, thereby placing GenAI models on the EU market, the GenAI model providers must comply with the relevant obligations outlined in the AI Act. To that end, GenAI model providers should establish a policy to comply with EU law on copyright and related rights, particularly to identify and respect the reservation of rights expressed by rights holders under Article 4(3) of the DSM Directive, regardless of the jurisdiction where the copyright-relevant acts underpinning the training of those GenAI models occur.[135] As explicitly stated in Article 2(1)(a) of the AI Act, "this AI Act applies to providers placing on the market or putting into service AI systems or placing on the market general-purpose AI models in the Union, irrespective of whether those providers are established or located within the Union or in a third country". Thus, the AI Act sets rules on exceptions or limitations to commercial text and data mining as market entry requirements for GenAI model providers entering the EU market. The European Parliament believes this is necessary to ensure a level playing field among providers of GenAI models, preventing any provider from gaining a competitive advantage in the EU market by applying lower copyright standards than those in the EU. If properly enforced, this extraterritorial application rule may protect rights holders in the EU.[136]

3.3.3. *Another example of the EU's rights-oriented approach: Chinese copyright law*

China has also adopted a rights-oriented approach to copyright in relation to GenAI model training. The Interim Measures for the Management of Generative Artificial Intelligence Services (IMMGAIS), which took effect in August 2023, regulates the handling of training data by GenAI service providers. It requires that GenAI service providers use training data and foundational models from legitimate sources.[137] In terms of intellectual property rights, GenAI service providers must ensure they do not infringe on the legally protected intellectual property rights of others.[138]

Article 24 of the Copyright Law of the People's Republic of China, along with Article 21 of its Implementing Regulations, outlines exceptions to copyright protection, allowing copyrighted works to be used in specific circumstances without the need for the copyright holders' consent or payment of royalties. These exceptions are contingent on the proper attribution of the author's name or the work's title, and on ensuring that the normal use of the work is not disrupted, nor are the legitimate rights and interests of the copyright holder unduly harmed. The exceptions encompass twelve explicitly defined situations, along with a catch-all clause for "other circumstances stipulated by laws and administrative regulations". However, these twelve situations do not extend to the use of copyrighted works for GenAI model training. Professor He Tianxiang has conducted an in-depth analysis of Article 24(1) and (6) of the Copyright Law concerning AI data training, concluding that these provisions are inapplicable to commercial AI data training.[139] He asserts that "if incomplete and biased data is fed into the AI system due to the impossible task of copyright clearance, the outcome (often in the form of prediction) will also be biased and even misleading. Therefore, there is an urgent requirement for countries to provide AI analysis and data mining at least with a shelter with its copyright exceptions or defences. Unfortunately, China has failed to accommodate this need".[140]

Regarding the use of copyrighted works in GenAI model training, the Chinese intellectual property law community has proposed establishing fair use rules.[141] Some scholars suggest creating machine learning fair use clauses to clarify the legality of using copyrighted works in GenAI

[132] Article 53(2), the AI Act.

[133] Claudio Novelli and others, 'AI Risk Assessment: A Scenario-Based, Proportional Methodology for the AI Act' (2024) 3 Digital Society; Claudio Novelli and others, 'Taking AI Risks Seriously: A New Assessment Model for the AI Act' [2023] AI & Society.

[134] Article 56(2)(b), the AI Act.

[135] Recital 106, the AI Act.

[136] Yan Wang, 'Do Not Go Gentle into That Good Night: The European Union's and China's Different Approaches to the Extraterritorial Application of Artificial Intelligence Laws and Regulations' (2024) 53 Computer Law & Security Review.

[137] Article 7(1), the IMMGAIS.

[138] Article 7(2), the IMMGAIS.

[139] Tianxiang He, 'Copyright Exceptions Reform and AI Data Analysis in China: A Modest Proposal' in Jyh-An Lee, Reto Hilty and Kung-Chung Liu (eds), *Artificial Intelligence and Intellectual Property* (Oxford University Press 2021).

[140] ibid.

[141] Jiyu Zhang and Saifei Wang, 'Research on Fair Use of Copyright in Large Model Data Training' (2024) 4 Journal of East China University of Political Science and Law 20; Yu Liu, 'Economic Analysis of Fair Use of Copyright by Machines Using Data' (2024) 3 Intellectual Property 107; Xiaochun Liu, 'Non-Work-Related Use in Generative AI Data Training and Its Legal Justification' (2024) 3 Law Forum 67; Xu Zhao, 'Issues of Fair Use in Machine Learning by Generative AI' (2024) 3 Jinan Journal (Philosophy and Social Sciences Edition) 79; An Li, 'Copyright Rules for Machine Learning: Historical Lessons and Contemporary Solutions' (2023) 6 Global Law Review 97.

model training.[142] Some advocate for defining data training as "non-work-related use", thereby excluding it from the rights conferred by copyright.[143] Additionally, some scholars argue that GenAI without a specific mimicry purpose could generally qualify as fair use, while GenAI designed to imitate specific works should not be considered fair use.[144] Furthermore, Fen Jiang and Hong Wu conducted an in-depth analysis of the judicial case *Shanghai Xinchuanghua Cultural Development Co., Ltd. v. AI Company (pseudonym)*, where the court provided preferential protection to the plaintiff's copyright.[145] Finally, regarding other issues related to GenAI and copyright, Chinese scholars have also engaged in-depth discussions on the copyright implications of content generated by GenAI.[146]

In the following part, we will demonstrate that both U.S. industry-oriented copyright law and EU rights-oriented copyright law have shortcomings in balancing the development of GenAI technology with copyright protection. Therefore, we need to seek better solutions.

4. Comparative analysis of U.S. and EU law and proposals for improvement

4.1. Insights from the tragedy of the commons and the tragedy of the anti-commons

We can draw inspiration from the tragedy of the commons[147] and the tragedy of the anti-commons[148] to adopt a more flexible approach to resolving the copyright issues in GenAI data training.

Garrett Hardin's famous article, "The Tragedy of the Commons", recognized the overuse that occurs when resources are freely available to everyone in common.[149] In tragedy of the commons, the resource which all have a liberty-right to use becomes a "commons", from which no one has a normative power to exclude others, and which no one has a duty to refrain from exploiting.[150] Where resources are freely available to everyone in common, everyone has an incentive to take as much as they desire, even though this collective behaviour may ultimately lead to the resource's destruction.[151] Hardin's depiction of the tragedy of the commons has served as fodder for intellectual property jurisprudence.[152]

With "tragedy of the commons" paradigm for intellectual property, Landes and Posner argue that the primary benefit of intellectual property rights is not that they generate incentives to create new works.[153] Empirically, it seems unlikely that we will witness a decrease in the diversity of available copyright works in a digital age of ever-reduced production and distribution costs and where the production of information has been characterized as following a "long tail" distribution.[154] Additionally, there has been no convincing evidence that piracy and infringement lead to fewer creative works. Instead, Landes and Posner assert that the most significant advantage of intellectual property rights lies in ensuring the efficient exploitation of existing intellectual works.[155] Professor Shubha Ghosh, from a different perspective, examines the application and adaptation of the "tragedy of the commons" concept within the intellectual property realm. He argues that, in this context, the "commons" issue is fundamentally about human creativity and innovation.[156]

Inspired by the concept of "the tragedy of the commons", we argue that if vast amounts of human copyright works can be freely used for GenAI model training by force of law, these works effectively become a "commons" available to GenAI model providers without restriction. The primary concern with this unrestricted use is the potential erosion of the principle that human creators are central to the copyright system, as well as the risk of diminishing the social core function and unique value of human-created works.[157] The solution to this tragedy in GenAI model training is not to rigidly uphold the exclusive rights of copyright holders — which could stifle derivative creativity in certain sectors[158] — but

[142] Zhang and Wang (n 141).
[143] Liu, 'Non-Work-Related Use in Generative AI Data Training and Its Legal Justification' (n 141).
[144] Zhao (n 141).
[145] Feng Jiang and Hong Wu, 'The World's First Completed Copyright Case Involving a Generative Artificial Intelligence Service Provider: Shanghai Xinchuanghua Cultural Development Co., Ltd. v. AI Company (Pseudonym)' (2024) 14 Queen Mary Journal of Intellectual Property.
[146] Haochen Sun, 'Redesigning Copyright Protection in the Era of Artificial Intelligence' (2022) 3 Iowa Law Review 107; Rita Matulionyte and Jyh-An Lee, 'Copyright in AI-Generated Works: Lessons from Recent Developments in Patent Law' (2022) 19 Scripted: A Journal of Law, Technology & Society 5; Tianxiang He, 'The Sentimental Fools and the Fictitious Authors: Rethinking the Copyright Issues of AI-Generated Contents in China' (2019) 27 Asia Pacific Law Review 218; Qian Wang, 'A Third Exploration of the Positioning of AI-Generated Content under Copyright Law' (2024) 3 Law and Business Research 182; Qian Wang, 'Revisiting the Classification of AI-Generated Content under Copyright Law' (2023) 4 Tribune of Political Science and Law 16; Qian Wang, 'On the Classification of AI-Generated Content under Copyright Law' (2017) 5 Science of Law (Journal of Northwest University of Political Science and Law) 148.
[147] Hardin (n 23).
[148] Heller (n 24).
[149] Hardin (n 23).
[150] Stephen R Munzer, 'The Commons and the Anticommons in the Law and Theory of Property' in Martin P Golding and Willaim A Edmundson (eds), *The Blackwell Guide to the Philosophy of Law and Legal Theory* (Blackwell Publishing 2005).
[151] Hardin (n 23); Elinor Ostrom, Roy Gardner and James Walker, *Rules, Games, and Common-Pool Resources* (The University of Michigan Press 1994); Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990); Barton H Thompson, 'Tragically Difficult: The Obstacles to Governing the Commons' (2000) 30 Environmental Law 241.
[152] W Landes and RA Posner, *The Economic Structure of Intellectual Property* (Harvard University Press 2003); SJ Liebowitz and SE Margolis, 'Seventeen Famous Economists Weigh in on Copyright: The Role of Theory, Empirics, and Network Effects' (2005) 18 Harvard Journal of Law and Technology 435; Lawrence Lessig, *The Future of Ideas: The Fate of the Commons in a Connected World* (Knopf Doubleday Publishing Group 2002); David E Adelman, 'A Fallacy of the Commons in Biotech Patent Policy' (2005) 20 Berkeley Technology Law Journal 985; Lee Anne Fennell, 'Common Interest Tragedies' (2004) 98 Northwestern University Law Review 907; Stewart E Sterk, 'Rhetoric and Reality in Copyright Law' (1996) 94 Michigan Law Review 1197; Shubha Ghosh, 'The Fable of the Commons: Exclusivity and the Construction of Intellectual Property Markets' (2007) 40 UC Davis Law Review 855; Tracy Reilly, 'Copyright and the Tragedy of the Common' (2014) 55 IDEA - The Intellectual Property Law Review 155.
[153] Landes and Posner (n 152).
[154] C Anderson, *The Long Tail* (Hachette Books 2008).
[155] Landes and Posner (n 152).
[156] Ghosh (n 152).
[157] Martin Senftleben, 'Generative AI and Author Remuneration' (2023) 54 IIC - International Review of Intellectual Property and Competition Law 1535; Christophe Geiger and Vincenzo Iaia, 'The Forgotten Creator: Towards a Statutory Remuneration Right for Machine Learning of Generative AI' (2024) 52 Computer Law & Security Review.
[158] Christophe Geiger, 'Statutory Licenses as Enabler of Creative Uses' in Kung-Chung Liu and Reto Hilty (eds), *Remuneration of Copyright Owners* (Springer 2017).

rather to maintain an anthropocentric approach to copyright law. This approach seeks to prevent a scenario where human creativity is overshadowed by AI tools.[159] We emphasize that the benefits of fostering the GenAI sector should always align with the goal of enhancing the freedom of human literary and artistic expression.[160]

GenAI is not a true creator. Its ability to mimic human literary and artistic expression comes solely from analyzing countless human creations used as training resources.[161] When GenAI uses vast amounts of human copyright works for model training without compensation, it can produce literary and artistic outputs more quickly and at lower costs, potentially chilling human creativity[162] and disrupting the market for human-created works.[163] Even more concerning is that as GenAI systems advance, human authors may face substitution effects due to the parasitic encroachment on the market for literary and artistic creation.[164] The foreseeable trend is that works generated by GenAI could eventually eliminate the demand for the very human creativity that provides GenAI its competitive edge.[165] As a result, human creators may lose income as machines replace them in fields ranging from journalism and writing to music and visual arts.[166] The unwritten right of human creators to receive remuneration for the commercial use of their works should be recognized as a fundamental and binding principle of copyright law, derived from the basic rights enshrined in international human rights conventions and the social function of copyright.[167] The parasitic use of human literary and artistic works by GenAI during training, without any compensation, undermines this right to fair remuneration, which is essential for encouraging the continued creation of new works by humans (including the expanded creative abilities that may eventually be enhanced through GenAI tools).[168] This could violate the principle that human creators are central to the copyright system.[169] Moreover, unlike works generated by GenAI, human literary and artistic creations serve a social core function and hold unique value for society as a whole.[170] Given the foreseeable trend that GenAI-generated works could ultimately eliminate the demand for the human creativity that gives GenAI its competitive edge,[171] copyright law in the GenAI era must continue to encourage human creation of new works to prevent the loss of the social core function and special value inherent in human creations.[172]

In contrast to the tragedy of the commons, the theory of the tragedy of the anticommons was first introduced by Michael Heller in his 1998 Harvard Law Review article. In this piece, he used the post-Soviet property system as an example to illustrate market failures caused by fragmented property rights and coordination breakdowns.[173] Professor Heller coined the term "tragedy of the anticommons" to describe "any setting in which too many people can block each other from creating or using a scarce resource".[174] As he explains, "the opposite of overuse in a commons is underuse in an anticommons".[175] Since then, the theory of the tragedy of the anticommons has been applied to the analysis of various intellectual property issues.[176] In the field of copyright law, this theory has been applied to analyze various issues, including joint authorship,[177] copyright divisibility,[178] fair use,[179] and copyright atomism.[180]

In the context of using vast amounts of copyrighted works for GenAI model training, if copyright law grants widely dispersed individual creators across various jurisdictions the rights to claim, retain, and manage their own copyrights, and if GenAI model providers are required to obtain authorization from each rights holder before training, the high transaction costs and complex authorization processes arising from fragmented copyrights and multiple ownerships across jurisdictions, could make it difficult for GenAI model providers to legally access the necessary human literary and artistic works for training. Since the continuous improvement of GenAI systems relies heavily on access to vast amounts of human literary and artistic creations, this could, in turn,

[159] Christophe Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' in F Thouvenin and A Peukert (eds), *Innovation - Creation - Markets* (Springer 2024).

[160] Geiger and Iaia (n 157).

[161] G Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' in M Garcia-Murillo, I MacInnes and A Renda (eds), *Handbook of Artificial Intelligence at Work: Interconnections and Policy Implications* (Edward Elgar 2023); G Frosio, 'Should We Ban Generative AI, Incentivise It or Make It a Medium for Inclusive Creativity?' in E Bonadio and C Sganga (eds), *A Research Agenda for EU Copyright Law* (Edward Elgar 2023); Geiger and Iaia (n 157); Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 159); Lemley and Casey (n 18).

[162] Geiger and Iaia (n 157); Sag (n 17).

[163] Senftleben, 'Generative AI and Author Remuneration' (n 157).

[164] ibid; Geiger and Iaia (n 157).

[165] Geiger and Iaia (n 157); Senftleben, 'Generative AI and Author Remuneration' (n 157).

[166] Senftleben, 'Generative AI and Author Remuneration' (n 157).

[167] Geiger and Iaia (n 157); Christophe Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' in Gustavo Ghidini and Valeria Falce (eds), *Reforming Intellectual Property Law* (Edward Elgar 2022).

[168] Senftleben, 'Generative AI and Author Remuneration' (n 157); Martin Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (2022); Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' (n 161); Frosio, 'Should We Ban Generative AI, Incentivise It or Make It a Medium for Inclusive Creativity?' (n 161); Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 159); Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167); FSE, 'Artificial Intelligence and the AI Act: A Joint Statement from Authors' and Performers' Organisations' (*Federation Screenwriters*, 2023).

[169] Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167); Geiger and Iaia (n 157); FSE (n 168).

[170] Martin Senftleben, *The Copyright/Trademark Interface - How the Expansion of Trademark Protection Is Stifling Cultural Creativity* (Kluwer Law International 2020); Senftleben, 'Generative AI and Author Remuneration' (n 157); Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (n 168); Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 159); Geiger and Laia (n 104).

[171] Geiger and Iaia (n 157); Senftleben, 'Generative AI and Author Remuneration' (n 157).

[172] Senftleben, 'Generative AI and Author Remuneration' (n 157).

[173] Heller (n 24).

[174] ibid; Michael Heller, 'The Tragedy of the Anticommons: A Concise Introduction and Lexicon' (2013) 76 Modern Law Review 6.

[175] Heller (n 145).

[176] Michael A Heller and Rebecca S Eisenberg, 'Can Patents Deter Innovation? The Anticommons in Biomedical Research' (1998) 280 Science 698; Reiko Aoki and Aaron Schiff, 'Promoting Access to Intellectual Property: Patent Pools, Copyright Collectives, and Clearinghouses' (2008) 38 R&D Management 189; Abraham Bell and Gideon Parchomovsky, 'Copyright Trust' (2015) 100 Cornell Law Review 1015; Jyh-An Lee, 'Copyright Divisibility and the Anticommons' (2016) 32 American University International Law Review.

[177] Bell and Parchomovsky (n 147).

[178] Lee (n 147).

[179] Ben Depoorter and Francesco Parisi, 'Fair Use and Copyright Protection: A Price Theory Explanation' (2002) 21 International Review of Law and Economics 453.

[180] Molly Shaffer and Van Houweling, 'Author Autonomy and Atomism in Copyright Law' (2010) 96 Virginia Law Review 549.

affect the quality of GenAI systems and hinder their development, as a significant portion of works might be excluded from the training pool.[181] As Llia Shumailov and co-authors noted in their *Nature* article, "AI models trained on AI-generated data descend into gibberish".[182] This echoes the old adage in information systems: "garbage in, garbage out".[183] Limited training datasets, whether due to technical or copyright restrictions, can marginalize the diversity of GenAI-generated outputs, increasing the likelihood of homogenization and bias.[184] Furthermore, given GenAI's inevitable presence in our daily lives and its potential to elevate human creativity to unprecedented levels through GenAI tools,[185] such limitations could stifle future creativity and undermine the very purpose of copyright: to encourage the creation and dissemination of works of authorship for the ultimate benefit of the public.[186]

By analyzing and evaluating the U.S. and EU laws from the standpoint of avoiding both the tragedy of the commons and the tragedy of the anti-commons, it is possible to identify a more balanced and effective institutional model.

4.2. Evaluation of U.S. law

Despite the foresight and reasonableness of non-expressive fair use in U.S. copyright law for addressing the copyright challenges posed by GenAI technology, there are notable shortcomings. This doctrine, by strictly distinguishing between creative expression and mere facts or ideas, effectively provides room for technological innovation while limiting direct copying that results in substantial similarity to the original works. This largely maintains a delicate balance between innovation and copyright protection. However, upon deeper analysis, it is evident that the existing U.S. legal framework still falls short in addressing the complexities of this emerging field, particularly in adequately considering economic compensation for copyright holders. This shortfall may exacerbate tensions within the creative industry in practice and potentially hinder the future creation and dissemination of artistic and scientific knowledge.

The crux of the issue lies in the fact that, while the doctrine of non-expressive fair use theoretically provides a legitimate basis for the training and learning processes of GenAI—allowing technology developers to utilize a vast number of existing works for algorithm optimization and model training without infringing on copyrights—this approach is essentially based on a "free-riding" logic. It overlooks the labor value of the original creators and their intangible contributions to the advancement of GenAI technology. One of the core purposes of copyright law is to incentivize creation by granting authors exclusive rights for a certain period, thereby promoting cultural and scientific prosperity. As mentioned earlier in this article, addressing the "tragedy of the commons" issue within the industry-oriented U.S. copyright law approach to GenAI model training is not about strictly upholding the exclusive rights of copyright holders, which could stifle derivative creativity in certain sectors.[187] Instead, the focus should be on maintaining an anthropocentric approach to copyright law that prevents a scenario where human creativity is overshadowed by GenAI tools.[188] Moreover, the solution is not to require GenAI model providers to obtain consent from every rights holder before training, as the high transaction costs and complex authorization processes could negatively impact the quality and development of GenAI systems.[189] Such a requirement could also stifle future creativity, ultimately undermining the core purpose of copyright: to encourage the creation and dissemination of works of authorship for the public's benefit.[190] Copyright law should aim to achieve significant public interest goals while developing new legal tools that benefit creators.[191] U.S. copyright law should aim for a balanced solution that reconciles copyright holders' demands for equitable remuneration with GenAI model providers' need for vast amounts of copyrighted works—ultimately fostering high-performance services that enhance human creativity while respecting the diverse interests protected by copyright law.[192]

However, when copyrighted works become indispensable "raw materials" for training GenAI models, the absence of a corresponding compensation mechanism within the legal system essentially contradicts this purpose. This could dampen the enthusiasm of creators, ultimately impacting the cultural creativity and knowledge output of society as a whole.

More importantly, as AI-generated content increasingly enters the market, its potential commercial value becomes more apparent, yet the contributions of the original copyrighted works are not adequately recognized economically. This not only challenges the fundamental fairness principles of copyright law but also prompts deep reflection on the relationship between technological development and social justice.

GenAI model training is analogous to the human learning process, as both draw on existing works to inform their creations. As long as the resulting expression is not substantially similar to existing works, there should be no copyright issues. So why should GenAI model training that uses existing works be condemned, especially when humans are not required to acknowledge every piece they have learned from when creating copyrighted works? We believe there is a strong case for this, based on the following four reasons.

Firstly, Christophe Geiger and Vincenzo Iaia argue that the right of human authors to receive remuneration for the commercial use of their intellectual efforts is protected under Article 27.2 of the Universal Declaration of Human Rights (UDHR) and Article 15.1(c) of the International Covenant on Economic, Social and Cultural Rights (ICESCR).[193] Conceptualizing copyright as a scientific and cultural right, a freedom of expression and artistic creation, and even as a socially rooted property clause should lead to the recognition of the unwritten right of human

[181] Ilia Shumailov and others, 'AI Models Collapse When Trained on Recursively Generated Data' (2024) 631 Nature 755; Gurol Canbek, 'Gaining Insights in Datasets in the Shade of "Garbage in, Garbage out" Rationale: Feature Space Distribution Fitting' (2022) 12 Data Mining and Knowledge Discovery 1.

[182] Shumailov and others (n 152).

[183] Canbek (n 152).

[184] Geiger and Iaia (n 157).

[185] Quintais (n 125).

[186] Geiger and Iaia (n 157).

[187] Geiger, 'Statutory Licenses as Enabler of Creative Uses' (n 158); Sean Flynn and Michael Palmedo, 'The User Rights Database: Measuring the Impact of Copyright Balance' (2019).

[188] Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 159).

[189] Shumailov and others (n 152); Canbek (n 152).

[190] Geiger and Iaia (n 157).

[191] Christophe Geiger and Oleksandr Bulayenko, 'Creating Statutory Remuneration Rights in Copyright Law: What Policy Options under the International Legal Framework?', *Intellectual Property Ordering Beyond Borders* (Cambridge University Press 2022).

[192] Christophe Geiger, 'Copyright as an Access Right, Securing Cultural Participation through the Protection of Creators' Interests' in R Giblin and KG Weatherall (eds), What if we could Reimagine Copyright? (Australian National University Press 2016); Christophe Geiger, 'Promoting Creativity through Copyright Limitations, Reflections on the Concept of Exclusivity in Copyright Law' (2010) 12 Vanderbilt Journal of Entertainment & Technology Law; JC Ginsburg, 'Fair Use for Free, or Permitted-but-Paid?' (2014) 29 Berkeley Technology Law Journal; A Kur, M Levin and J Schovsbo, 'Expropriation or Fair Game for All? The Gradual Dismantling of the IP Exclusivity Paradigm' in A Kur and M Levin (eds), Intellectual Property in a Fair World Trade System - Proposals for Reforming TRIPS (Edward Elgar 2011).

[193] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).

creators to receive remuneration for the commercial use of their works.[194] This right should be understood as a fundamental and binding principle of copyright law, derived from basic rights and the social function of copyright.[195] GenAI only acquires the ability to mimic human literary and artistic works after extracting patterns from countless human creations used as training data.[196] The parasitic use of human literary and artistic works during GenAI model training, without any compensation, deprives human authors of this right to fair remuneration under the UDHR and the ICESCR. This could violate the principle that human creators are central to the copyright system.[197]

Secondly, unlike works generated by GenAI, human literary and artistic creations serve a core social function and hold unique value for society.[198] In the GenAI era, copyright law must continue to encourage human creativity, including the expanded creative abilities that might eventually be facilitated by GenAI, to prevent the loss of the social core function and unique value of human works.[199] In this context, copyright measures that provide remuneration to human authors who supply the raw material for GenAI model training are a fitting solution. First, human literary and artistic expression can reflect the flaws of contemporary society, expose the shortcomings of existing social and political conditions, and open new horizons for social development, acting as a crucial catalyst for social and political change.[200] Unlike humans, GenAI systems cannot experience or be influenced by contemporary social conditions, and therefore cannot inspire visions of new moral norms that resonate with current societal aspirations.[201] Second, human literary and artistic works possess inherent social value. Aesthetic play is a source of human happiness, moral and political cultivation, imaginative freedom, and self-realization. If aesthetic play is relegated to GenAI, humans lose the opportunity to experience happiness, moral and political cultivation, imaginative freedom, and self-realization within society.[202] Lastly, human avant-garde movements lead to new forms of expression and new reflections on social conditions.[203] GenAI struggles to independently introduce disruption into the established order, reveal societal tensions and conflicts, and propose transformative ideas through avant-garde works.[204]

Thirdly, drawing an analogy to the concept of *domaine public payant*, the proposal to charge GenAI model providers as a means of compensating human creators is a reasonable approach within copyright law. Professor Adolf Dietz introduced the *domaine public payant* concept, arguing that copyright law should address "the existential and financial misery of an artist's life".[205] As GenAI systems evolve, human authors are likely to face substitution effects,[206] potentially losing income as GenAI replaces them in areas such as journalism, writing, music, and visual arts.[207] Professor Martin Senftleben highlights a striking similarity between the concept of *domaine public payant* and the idea of charging GenAI.[208] He observes, "Both concepts concern creations that fall outside the scope of individual authors' exploitation rights: literary and artistic works that never enjoyed or no longer enjoy copyright protection in the case of *domaine public payant*, and general ideas, concepts, and styles in the case of AI output that does not reproduce any individual expression of a human author. At the same time, it is clear that both concepts pertain to literary and artistic subject matter: public domain works, and public domain ideas, concepts, and styles".[209]

Finally, the continuous improvement of GenAI depends on human literary and artistic creations. In an article published in *Nature*, Llia Shumailov and co-authors found that "AI models trained on AI-generated data descend into gibberish".[210]

In the absence of effective compensation mechanisms, copyright holders often face a dilemma: either accept that their works are used without compensation to train GenAI models that may generate substantial commercial benefits, or enforce strict copyright protection measures, which could impede technological progress and the maximization of public benefits.

Therefore, while the reasonableness of non-expressive fair use in U.S. copyright law is commendable, it does not fully address the issue of economic compensation for copyright holders. To achieve a win-win situation for technological innovation and copyright protection, it is necessary to introduce innovative solutions. These could include establishing a dedicated compensation fund, devising fair royalty distribution schemes, or exploring other incentive-compatible mechanisms. Such measures would ensure that, while fostering innovation, the creative labor of copyright holders is also respected and rewarded, thereby laying a more solid and just legal foundation for the sustainable development of GenAI.

4.3. Evaluation of EU law

The EU legal framework has demonstrated unprecedented foresight and standardization in addressing the rapid development of GenAI

[194] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).

[195] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).

[196] Lemley and Casey (n 18).

[197] Senftleben, 'Generative AI and Author Remuneration' (n 157); Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (n 168); Frosio, 'Should We Ban Generative AI, Incentivise It or Make It a Medium for Inclusive Creativity?' (n 161); Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' (n 161); Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167); FSE (n 168).

[198] Senftleben, 'Generative AI and Author Remuneration' (n 126); Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (n 137); Senftleben, The Copyright/Trademark Interface - How the Expansion of Trademark Protection Is Stifling Cultural Creativity (n 140); Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' (n 130); Frosio, 'Should We Ban Generative AI, Incentivise It or Make It a Medium for Inclusive Creativity?' (n 130); Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 128); Geiger and Iaia (n 126).

[199] Senftleben, 'Generative AI and Author Remuneration' (n 157).

[200] ibid; B Beebe, 'Bleistein, the Problem of Aesthetic Progress, and the Making of American Copyright Law' (2017) 117 Columbia Law Review 319.

[201] Martin Senftleben and L Buijtelaar, 'Robot Creativity: An Incentive-Based Neighbouring Rights Approach' (2020) 42 European Intellectual Property Review 797.

[202] Beebe (n 171).

[203] Senftleben, 'Generative AI and Author Remuneration' (n 157); Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (n 168).

[204] Senftleben, 'Generative AI and Author Remuneration' (n 157); Senftleben, 'A Tax on Machines for the Purpose of Giving a Bounty to the Dethroned Human Author - Towards an AI Levy for the Substitution of Human Literary and Artistic Works' (n 168).

[205] A Dietz, 'A Modern Concept for the Right of the Community of Authors (Domaine Public Payant)' (1990) 24 Copyright Bulletin 13.

[206] Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' (n 161); Senftleben, 'Generative AI and Author Remuneration' (n 157).

[207] Senftleben, 'Generative AI and Author Remuneration' (n 157).

[208] ibid.

[209] ibid.

[210] Shumailov and others (n 152).

technology, especially in the field of copyright law. Specifically, its requirement for GenAI model providers to explicitly disclose the use of copyrighted materials in their training data is a significant step toward establishing a solid foundation for compensation mechanisms and protecting the rights of original creators. By enhancing transparency, EU law forces GenAI model developers to consider and report the use of copyrighted work they employ, promoting compliance with copyright laws and providing copyright holders with a means to monitor the use of their works. This transparency provides essential data and practical support for developing specific compensation measures in the future. In the long run, this regulation helps build a fairer, more transparent GenAI ecosystem, ensuring the harmonious coexistence of innovation and intellectual property protection.

However, the rights-oriented EU copyright law has its shortcomings, particularly the provision allowing copyright holders to make specific reservations to prohibit the use of their works for GenAI training. While this regulation aims to strengthen the control of copyright holders and ensure their works are not used against their will, reflecting a respect for personal creative freedom and rights, its potential negative impacts cannot be overlooked.

First, this regulation does not align with the vast data demands for GenAI model training. Given that copyright holders have incentives to exercise the right to reservation under rights-oriented EU copyright law, the exercise of this right fundamentally does not differ from the "prior consent" model, where the use of copyrighted works requires prior consent from the copyright holder. Although the use of copyrighted works based on the right to reservation and prior consent essentially follows the logic of property law—granting copyright holders the opportunity and ability to negotiate, enabling the effective circulation of intellectual works in the market, and promoting the production of innovative expressions and the sharing of knowledge—GenAI model training often requires vast amounts of data. The copyright transaction model of "right to reservation, prior consent, and royalties" cannot meet the massive data learning needs of the GenAI era. This conflict easily leads to low transaction efficiency, increased transaction costs, and ineffective GenAI data training.

Regarding the authorization process for GenAI model training, GenAI model providers, as data demanders, need to obtain prior consent from copyright holders who have exercised the right to reservation to avoid potential infringement risks. However, obtaining authorization for related works is not a simple process of "issuing an offer and reaching an agreement". It often involves the identification costs of data acquisition and the negotiation costs of data transactions. In terms of identification costs, the operation of GenAI requires massive data support, much of which comes from copyrighted works, especially high-quality data content which is mostly copyrighted. However, collecting data from copyrighted works requires identifying the source and ownership of the works and accurately locating the true copyright holder, which undoubtedly imposes a significant transaction burden on GenAI service providers. Furthermore, in terms of negotiation costs, after determining the works to be collected and the copyright holders, it is also necessary to negotiate the price and authorization scope of the data acquisition. If the collected data content has unclear ownership or sources, the transaction costs for GenAI service providers will undoubtedly rise, making it difficult to meet the data training needs of GenAI.

Lastly, this regulation might lead to biases and discrimination issues in GenAI models due to the inability to access comprehensive and diverse datasets during training. In an era where GenAI technology heavily relies on extensive data learning, the breadth and diversity of data are crucial for ensuring the accuracy and broad applicability of models. Although well-intentioned, the copyright holders' reservation rights could create data silos, limiting GenAI algorithms' ability to learn from specific types of information, thereby affecting the fairness and effectiveness of the models. The ripple effects of this approach extend beyond the technical realm and touch on deeper considerations of social ethics and justice. Model bias may exacerbate existing social inequalities. For example, in scenarios such as recruitment, credit evaluation, and legal judgments, decision-making biases due to insufficient training data could adversely affect specific groups, hindering the potential of GenAI technology as a tool for public good. Additionally, it may slow down the progress of GenAI technology, as true GenAI must be based on a comprehensive, unbiased understanding of the world. Any limitations in data will pose significant obstacles to its development. Therefore, while EU law should fully protect the rights of creators, it should not hinder the reasonable flow and use of data content. It is crucial to avoid obstructing GenAI technological innovation while striving to build a future that respects knowledge creation and promotes technological inclusivity.

4.4. Propositions: fair remuneration rights for GenAI training under the UN's international governance regime for AI

4.4.1. General discussion

In the face of the copyright law challenges posed by the development of GenAI models, national legislation alone is insufficient and does not support the widespread global application of GenAI technology. It is therefore ideal for different jurisdictions to collaborate and address these challenges by formulating binding or non-binding norms at the global level. As mentioned in Section 3.1 of this article, the United Nations, at the heart of the rules-based international order, is leading efforts to build an international governance regime for AI.[211] This provides the institutional and normative foundation for collective action in the global governance of AI. The report titled "Governing AI for Humanity", released by the UN Advisory Body on Artificial Intelligence, indicates that the United Nations has incorporated the issue of balancing GenAI model training with copyright protection into the UN-led international governance regime for AI.[212] The UN High-Level Advisory Body on AI believes that a core purpose of global AI governance is to address the risks associated with AI, including "the impact on labor markets if large sections of the workforce are displaced, or on creativity if intellectual property rights are not protected".[213] In this context, the UN plans to conduct an in-depth discussion on the intellectual property aspects of global AI governance.[214] The importance the UN places on balancing GenAI innovation with copyright protection is also reflected in two AI-related resolutions adopted by the UN General Assembly in 2024. These resolutions emphasize the importance of "encouraging, where appropriate and relevant, the implementation of appropriate safeguards to respect intellectual property rights, including copyright-protected content, while promoting innovation", and that "Member States should enjoy equal opportunities in the use of artificial intelligence, while respecting intellectual property rights and promoting innovation".[215]

This article proposes the creation of a pathway within the UN-led international AI governance regime to balance GenAI model training with copyright protection. The UN acknowledges that within this international governance framework, "We cannot rule out that legally binding norms and enforcement would be required at the global level. […] Non-binding norms could also play an important role, alone or in combination with binding norms". Thus, the approach to balancing GenAI model training with copyright protection at the international level—whether through binding norms, non-binding norms, or a combination of both—will largely depend on the complex political dynamics among member states. From the perspective of international copyright

[211] United Nations, 'Interim Report: Governing AI for Humanity' (n 43).
[212] ibid.
[213] ibid.
[214] ibid.
[215] United Nations, 'Resolution Adopted by the General Assembly on 21 March 2024: 78/265. Seizing the Opportunities of Safe, Secure and Trustworthy Artificial Intelligence Systems for Sustainable Development' (n 44).

standards, even though the political climate around establishing further international copyright law standards is sensitive, a second "Miracle of Marrakesh" could still emerge in the field of GenAI. Given the uncertainty surrounding this issue, this article refrains from advocating for binding norms as the sole solution to balancing GenAI model training with copyright protection at the international level. Instead, it suggests following the UN-led international governance regime for AI by setting timeframes for the realization of subfunctions and implementing them in phases. According to the UN's plan, concerning the subfunction of policy harmonization and norm alignment, the UN intends to, under its international governance regime for AI, "surface best practices for norms and rules, including for risk mitigation and economic growth. Align, leverage, and include, soft and hard law, standards, methods, and frameworks developed at the regional, national, and industry level to support interoperability".[216] Therefore, this article advocates for incorporating the development and implementation of both binding and non-binding norms aimed at mitigating the risks to copyright and job displacement caused by GenAI, as well as promoting GenAI innovation and development, within the UN's international governance regime for AI.

Regarding the institution(s) specifically responsible for developing guidelines, norms, and rules to balance GenAI model training with copyright protection, the WIPO appears to be a suitable choice. As the UN agency for intellectual property, WIPO provides a policy forum to shape balanced international IP rules for a changing world and offers global services to protect IP across borders and resolve disputes. However, given WIPO's tendency to move slowly, this article does not foresee binding norms for balancing GenAI innovation and copyright protection being agreed upon among member states in the near future. A more practical approach might be to first build a broad consensus on the issue of GenAI model training and copyright protection through the annual WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI). As discussed in Section 3.1 of this article, WIPO addressed the topic of equitable remuneration during its second session. The WIPO paper titled "Revised Issues Paper on Intellectual Property Policy and Artificial Intelligence", released during that session, raises the question: "Should remedies for infringement be limited to equitable remuneration?"[217] Once a broad consensus is achieved, norms balancing GenAI model training with copyright protection—whether non-binding or binding—could then be proposed, depending on the political dynamics among countries.

As discussed in Sections 4.1 and 4.2 of this article, drawing on insights from the tragedy of the commons and the tragedy of the anticommons, one viable solution to address the shortcomings of both U.S. industry-oriented copyright law and EU rights-oriented copyright law in balancing the development of GenAI technology with copyright protection is to provide fair remuneration to copyright holders whose works are used without compensation in GenAI model training. Since the UN-led international governance regime for AI is grounded in the UN Charter, international human rights law, and other internationally agreed-upon commitments,[218] the solution to the copyright protection issue in GenAI model training should also be based on these principles. As analyzed in Section 4.2 of this article, the right to fair remuneration is recognized as an international human right. Christophe Geiger and Vincenzo Iaia argue that the right of human authors to receive remuneration for the commercial use of their intellectual efforts is protected under Article 27.2 of the UDHR and Article 15.1(c) of the ICESCR.[219] This right should be understood as a fundamental and binding principle of copyright law, rooted in basic rights and the social function of copyright.[220] The parasitic use of human literary and artistic works during GenAI model training, without any compensation, deprives human authors of this right to fair remuneration under the UDHR and the ICESCR, potentially violating the principle that human creators are central to the copyright system.[221]

It is important to note that when it comes to coordinating copyright exceptions for commercial GenAI model training at the international level, most international copyright treaties primarily leave the determination of exceptions to national legislatures and courts.[222] As analyzed in Section 3.1 of this article, Article 9(2) of the Berne Convention assigns the responsibility for establishing limitations and exceptions to copyright protection to the domestic laws of member states, with the mandatory condition of adhering to the "three-step test".[223] Subsequent intellectual property treaties have largely preserved this approach to limitations and exceptions, as set forth by the Berne Convention.[224] While there are precedents for international treaties mandating the global implementation of uniform copyright exceptions—such as the Marrakesh Treaty—this article suggests that, given the complex political environment surrounding copyright exceptions in the current international copyright law landscape, it may be more practical to continue leaving the determination of copyright exceptions for GenAI model training to national legislatures and courts.

4.4.2. Typology of specific models for the right to fair remuneration for GenAI training

This section offers a typological analysis of potential operational models for the right to fair remuneration for GenAI training, drawing analogies to existing types of remuneration rights in international copyright treaties. The goal is to provide an open framework for further discussion within the international academic community.

4.4.2.1. A collective remuneration right per se for GenAI training beyond the minimum scope provided by international copyright treaties: an analogy to domaine public payant. Remuneration rights *per se* for GenAI training refer to rights to remuneration established by international treaties, regional norms, or national legislation outside the scope of those treaties. The creation of such rights does not introduce exceptions or limitations to the exclusive rights provided by international copyright treaties.[225]

As discussed in Section 4.2 of this article, the proposal to charge GenAI model providers as a means of compensating human creators, drawing an analogy to the concept of *domaine public payant*, may be a reasonable approach within copyright law. Professor Adolf Dietz introduced the *domaine public payant* concept, suggesting that in addition to the traditional rights of individual authors to use and be remunerated for their works, it is both reasonable and desirable to recognize a new right within copyright law. This right would be collectively enjoyed by a

[216] United Nations, 'Interim Report: Governing AI for Humanity' (n 43).
[217] WIPO, 'WIPO Conversation on Intellectual Property (IP) and Artificial Intelligence (AI): Revised Issues Paper on Intellectual Property Policy and Artificial Intelligence' (n 46).
[218] United Nations, 'Interim Report: Governing AI for Humanity' (n 43).
[219] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).
[220] Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167); Geiger and Iaia (n 157).
[221] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).
[222] Ricketson and Ginsburg (n 58); SM Fiil-Flynn and others, 'Legal Reform to Enhance Global Text and Data Mining Research: Outdated Copyright Laws around the World Hinder Research' (2022) 378 Science 951.
[223] Ricketson and Ginsburg (n 58).
[224] ibid.
[225] Geiger and Bulayenko (n 191).

different rights holder—the "community of authors".[226] He explained, "What we finally propose is simply to introduce another rights owner, namely the community of living and creating authors, among several kinds of rights owners already existing in copyright law. This community of authors should have the direct right to participate in the income from the exploitation of works of deceased authors after the individual term of copyright protection has expired".[227]

Existing international copyright treaties do not mandate remuneration for the use of works whose protection term has expired, as they only establish the minimum protection period. Currently, only a few countries have enacted laws establishing *domaine public payant*. The issue of *domaine public payant* continues to be a recurring topic in national policy debates. For example, Article 59(1) of Annex VII to the Bangui Agreement introduces the concept of *domaine public payant*. Under Article 59 (2) of Annex VII, the payment for the use of public domain works is to be set at half the usual amount specified by contracts and practices for the use of protected works.

The legal rationale behind the *domaine public payant* concept is to close the "harmful gap" between the substantial profits made by those who exploit public domain works and the precarious working and living conditions of the original authors.[228] Similarly, as GenAI advances, human authors are likely to experience substitution effects, potentially losing income as GenAI replaces them in fields such as journalism, writing, music, and visual arts.[229]

Drawing an analogy to the *domaine public payant* concept, member states could legislate to establish a collective remuneration right *per se* for GenAI training, extending beyond the minimum remuneration rights provided by international copyright treaties. The enactment of such a right would not create exceptions or limitations to the exclusive rights established by these treaties. Instead, the exceptions or limitations to copyright protection related to GenAI training would be determined by the domestic legislation of each member state. Thus, the collective remuneration right *per se* for GenAI training, which goes beyond the minimum scope of international copyright treaties, would need to be integrated with domestic copyright exception rules to achieve a balance between the benefits of GenAI training and copyright protection.

Under the model of a collective remuneration right *per se* for GenAI training, which extends beyond the minimum remuneration rights provided by international copyright treaties, member states could legislate to establish this collective right specifically for GenAI training. As a potential design, GenAI model providers could be required to pay fair collective remuneration to one or more copyright collective management organizations designated by national legislation. The rules governing these payments could also be established through national legislation. The remuneration rate could be negotiated by the collective management organizations or set by legislation, with a designated administrative body determining the specific standards. During negotiations, collective management organizations could consider implementing tiered remuneration rates based on the varying magnitudes of data usage, the specific depth of data mining, and the quality of data content required for GenAI model training. This would naturally fall within the scope of private law negotiation. The level of fair remuneration could be linked to a percentage of the GenAI model providers' expenditures or revenue. This approach ensures that as industry revenues grow, remuneration increases proportionately; conversely, if industry revenues decline, the financial burden on the GenAI sector would be reduced. This collective remuneration right could be structured either as a one-time payment based on specific criteria or as a recurring payment, rather than a single, one-off charge. The collective management organizations designated by national legislation would be responsible for developing a distribution plan, establishing specific funds (such as a Human Creativity Promotion Fund), and managing and disbursing the collected funds. These funds would be used to improve the working and living conditions of human authors and to promote and support human literary and artistic creation. The distribution plan of the collective management organization could include provisions for direct payments to authors, ensuring that funds reach not only prominent star authors but also individual creators. This model for a GenAI remuneration right has the advantage of avoiding the "tragedy of the anticommons", as discussed in Section 4.1 of this article. It does not require GenAI model providers to identify every specific rights holder of the works used in GenAI training, nor does it necessitate the construction of a complex and costly rights clearance infrastructure.

4.4.2.2. A limitation-based remuneration right for GenAI training within the flexibility of the "three-Step test". Unlike remuneration rights *per se* for GenAI training, limitation-based remuneration rights for GenAI training involve converting the exclusive rights provided by international norms into remuneration rights under specific conditions permitted by the treaties.[230] In existing international copyright treaties, many exceptions and limitations are explicitly mentioned. Some articles require remuneration to be paid to copyright holders, while others do not. Regardless of whether remuneration is required, these articles can be implemented in national or regional laws as exceptions and limitations to exclusive rights. Other exceptions and limitations to exclusive rights, not explicitly mentioned in international copyright treaties, are created through the legislative flexibility allowed by the "three-step test".[231]

Limitation-based remuneration rights for GenAI training help reconcile the interests of GenAI model providers with those of human creators regarding fair compensation. This limitation requires GenAI model providers to fulfill a general remuneration payment obligation to human creators when using copyrighted works for model training purposes.[232] The introduction of such a right supports the anthropocentric approach to copyright law discussed in Sections 4.1 and 4.2 of this article, aiming to prevent a scenario where human creativity is overshadowed by GenAI tools.[233] This approach maximizes the availability of copyrighted content for GenAI training while ensuring that authors receive compensation for the commercial use of their intellectual efforts. As analyzed in Section 4.2 of this article, Christophe Geiger and Vincenzo Iaia argue that the right to fair remuneration is protected under Article 27.2 of the UDHR and Article 15.1(c) of the ICESCR.[234] Consequently, GenAI model providers should share their revenues with the authors whose works are used to train these models. A limitation-based remuneration right for GenAI training encourages the continued creation of new works, which may ultimately benefit from the expanded creative capabilities offered by GenAI, while ensuring that the use of their work by GenAI generates a fair return. In this way, human creators remain central to the copyright system.[235] Introducing such a right for

[226] Dietz (n 205).
[227] ibid.
[228] ibid.
[229] Frosio, 'The Artificial Creatives: The Rise of Combinatorial Creativity from Dall-E to GPT-3' (n 161); Senftleben, 'Generative AI and Author Remuneration' (n 157).
[230] Geiger and Bulayenko (n 191).
[231] ibid.
[232] Frosio, 'Should We Ban Generative AI, Incentivise It or Make It a Medium for Inclusive Creativity?' (n 161).
[233] Geiger, 'When the Robots (Try to) Take Over: Of Artificial Intelligence, Authors, Creativity and Copyright Protection' (n 159); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).
[234] Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167); Geiger and Iaia (n 157).
[235] Geiger and Iaia (n 157); Geiger, 'Building an Ethical Framework for Intellectual Property in the EU: Time to Revise the Charter of Fundamental Rights' (n 167).

GenAI training for commercial purposes could be a sound solution to protecting the threatened remuneration rights of creators without excessively compromising the ability of GenAI model providers to offer high-performance services that enhance human creativity.

The establishment of limitation-based remuneration rights for GenAI training can take one of the following three forms: a limitation-based remuneration right for GenAI training created solely under the flexibility of the "three-step test", a limitation-based remuneration right for GenAI training provided directly by international treaties, or a limitation-based remuneration right for GenAI training established within the scope of non-remunerative exceptions and limitations outlined by international treaties.

For a limitation-based remuneration right for GenAI training created solely under the flexibility of the "three-step test", countries can introduce such a right, which is not explicitly mentioned in copyright treaties, within the scope allowed by international legal frameworks. This can be achieved by leveraging the flexibility provided by the "three-step test", as outlined in Article 13 of the TRIPS Agreement, Article 9(2) of the Berne Convention, Article 10(1) of the WCT, Article 16(2) of the WPPT, and Article 13(2) of the Beijing Treaty.

The flexibility of the "three-step test" lies precisely in its ambiguity and the absence of clear boundaries.[236] This flexibility has also contributed to its successful integration into copyright legislation. On the question of whether the "three-step test" should be applied sequentially or as a whole, Professors Geiger and Senftleben offer an authoritative explanation: the purpose of the "three-step test" is to ensure an appropriate balance between the different interests involved.[237] Therefore, whether these steps are followed in sequence or not, the "three-step test" always functions as a single, integrated analysis.[238] Furthermore, providing fair remuneration to copyright holders under exceptions and limitations is more likely to satisfy the "three-step test" requirement of preventing unreasonable prejudice to the rights holders, compared to non-remunerative exceptions or limitations.

Consider the example of private copying: although international treaties do not explicitly permit the copying of works for private use, the remuneration right for private copying, which substitutes for the corresponding exclusive right, is a typical example of a limitation-based remuneration right created under the flexibility of the "three-step test". This statutory mechanism provides a source of income for rights holders by imposing fees on manufacturers, importers, and/or retailers of private copying devices, as well as on economic operators who supply these devices to individuals. While rights holders cannot control the sale of copying-capable devices, they are entitled to remuneration. Many countries have adopted limitation-based remuneration rights for private copying. For example, in the EU, Article 5.2(b) of the Information Society Directive introduces a remunerated private copying exception or limitation mechanism, which member states can use to compensate rights holders for the losses caused by the private copying of their works. Similarly, in the case of GenAI model training, copyright holders cannot monitor or enforce their rights while their works are used without compensation. The U.S. copyright system has developed a "permitted-but-paid" regime, which limits the exclusive rights of copyright holders for specific purposes, such as preventing monopolies in the music industry and reducing licensing transaction costs for audiovisual works and television programs.[239]

Introducing a limitation-based remuneration right for GenAI training, created solely under the flexibility of the "three-step test", offers a sound solution for protecting the threatened remuneration rights of creators while allowing GenAI providers to continue delivering high-performance services that enhance human creativity. This approach carries no risk of violating the requirements of the "three-step test". Indeed, limitation-based remuneration rights for GenAI training are generally likely to meet the criteria set forth in Article 9 of the Berne Convention and Article 13 of the TRIPS Agreement.[240]

Regarding the specific operational design of the limitation-based remuneration right for GenAI training, created solely under the flexibility of the "three-step test", two options can be considered. One option is a private law approach, where the remuneration rate is negotiated and determined through collective management organizations. Member states could designate an agency specifically responsible for copyright and digital services to assist and facilitate these negotiations.[241] Alternatively, a public integration approach could be adopted, where the parameters for determining the remuneration rate are defined by statutory law. Member states could establish the standards for quantifying limitation-based remuneration rights for GenAI training through legislation or a designated administrative agency. When designing the specific operation of this system, requiring GenAI model providers to identify every rights holder and compensate them through an effective rights clearance infrastructure would inevitably result in high transaction costs and burdens. This, as analyzed in Section 4.1 of this article, could result in the "tragedy of the anti-commons", ultimately undermining the entire remuneration effort. A useful reference could be the levy system established under Germany's 1965 Copyright Act.[242] In this system, payments are made by imposing fees on the cost of physical media or devices that allow users to copy copyrighted works, and the collected fees are then distributed to content creators and rights holders according to various criteria among the member states implementing Article 5.2(b).[243] In the context of GenAI training, this limitation-based remuneration right could be based on the data used for machine learning with copyrighted works for commercial GenAI training purposes. The level of fair remuneration could also be benchmarked to a certain percentage of GenAI model providers' expenditures or revenues, as discussed in Section 4.4.2.1 of this article. This approach ensures that remuneration increases proportionally as industry revenues grow, while also preventing an excessive burden on the GenAI sector if revenues decline.

4.4.2.3. Other types of fair remuneration rights for GenAI training. The model for fair remuneration rights for GenAI training can also encompass other types, such as a fair remuneration right *per se* for GenAI training as provided by international treaties, a remuneration right *per se* for GenAI training that involves the coexistence and overlap of exclusive rights provided by international treaties, a limitation-based remuneration right for GenAI training under international treaties, and a limitation-based remuneration right for GenAI training created within the scope of non-remunerative exceptions and limitations established by international treaties.

First, we can explore a fair remuneration right *per se* for GenAI training provided by international treaties by drawing an analogy to the fair remuneration rights for broadcasting and public communication. Article 12 of the Rome Convention and Article 15(1) of the WPPT establish the right to fair remuneration for the use of commercial phonograms in broadcasting or any public communication. These remuneration rights are provided by international treaties and typically do not correspond to an equivalent exclusive right. Most countries have

[236] Geiger and Bulayenko (n 191).
[237] Geiger and Iaia (n 157); Senftleben, 'Generative AI and Author Remuneration' (n 157).
[238] Senftleben, 'Generative AI and Author Remuneration' (n 157); Geiger and Iaia (n 157).
[239] Ginsburg (n 192).
[240] Christophe Geiger, Daniel Gervais and Martin Senftleben, 'The Three-Step Test Revisited: How to Use the Test's Flexibility in National Copyright Law' (2014) 29 American University International Law Review 581.
[241] Geiger and Iaia (n 157).
[242] Martin Kretschmer, 'Private Copying and Fair Compensation: An Empirical Study of Copyright Levies in Europe' (2011).
[243] Ginsburg (n 192).

implemented these remuneration rights. However, since the Rome Convention and WPPT allow contracting parties to exclude or partially apply these rights through formal notification, the scope of these rights can vary significantly between countries. Similarly, the fair remuneration right for GenAI model training is intended to balance the interests of rights holders and users of copyrighted works—specifically, GenAI model providers. It is not an exception or limitation to an equivalent exclusive right. This fair remuneration right is triggered only when copyrighted works are directly or indirectly used by GenAI for model training. The assessment of whether the payment is "fair" should consider the value of the relevant use in GenAI model training, or even the value of the GenAI output. However, unlike the fairness assessment for broadcasting and public communication, where the focus is on the value of individual works, the copyrighted works used in GenAI model training are vast in number, making it extremely difficult, if not practically impossible, to assess the value of a single, specific copyrighted work.

Second, compared to the two types of remuneration rights *per se* for GenAI training discussed earlier in this article, designing a remuneration right *per se* for GenAI training that coexists and overlaps with the exclusive rights provided by international treaties appears unnecessary. In practice, the Beijing Treaty is the only international multilateral treaty that explicitly mentions a remuneration right *per se* that coexists and overlaps with the exclusive rights provided by international treaties. Article 12(3) of the Beijing Treaty states, irrespective of the transfer of exclusive rights, national laws may provide performers with the right to receive royalties or equitable remuneration for any use of their performances as set out in this Treaty, including the rights provided for in Articles 10 and 11. This article of the Beijing Treaty clearly establishes the possibility of maintaining remuneration rights even after the transfer of exclusive rights. It demonstrates that, although countries can grant a remuneration right that coexists and overlaps with the exclusive rights provided by international treaties through domestic law, the primary reason for granting this remuneration right is to address the issue of distributive justice for authors and performers when they transfer their exclusive rights to audio and/or audiovisual producers. However, in the context of GenAI model training, the exclusive rights to the author's works are not transferred to the GenAI model providers. Therefore, this type of remuneration right appears to have limited relevance when designing fair remuneration for GenAI training.

Finally, we could also consider adopting a limitation-based remuneration right for GenAI training provided by international treaties. For example, Article 11bis(2) of the Berne Convention grants contracting parties the authority to determine the conditions under which authors may exercise the rights provided in this article, provided that it does not affect the author's right to equitable remuneration. Similarly, Article 13 (1) of the Berne Convention allows each country of the Union to impose reservations and conditions on the exclusive right granted to the authors of authorized recordings of musical works, as long as equitable remuneration is ensured. The Marrakesh Treaty stipulates that contracting parties may establish remuneration for certain exceptions and limitations. Additionally, within the scope of non-remunerative exceptions and limitations provided by international treaties, a limitation-based remuneration right for GenAI training could also be established. The articles in international treaties that foresee the introduction of non-remunerative exceptions and limitations can serve as a foundation for introducing a limitation-based remuneration right for GenAI training.[244] Examples include Article 15(2) of the Rome Convention, Article 6 of the Phonograms Convention, Article 16(1) of the WPPT, and Article 13(1) of the Beijing Treaty, which contain general articles allowing contracting parties to provide exceptions or limitations to the rights of performers, audio and audiovisual producers, and broadcasting organizations similar to those applicable to copyright. When designing a fair remuneration system under this model, reference can be made to the limitation-based remuneration right for GenAI training created under the flexibility of the "three-step test" discussed in Section 4.4.2.2 of this article. As noted in Section 4.4.1 of this article, establishing new international copyright treaties in the current complex and sensitive international political environment is extremely difficult. However, under the UN-led international governance regime for AI and with WIPO's coordination, the establishment of a limitation-based remuneration right for GenAI training provided by international treaties could potentially become the next "Miracle of Marrakesh" (though highly improbable).

5. Conclusion

Although the rapid development of GenAI technology has brought unprecedented opportunities to society, its use of training data touches the core of copyright protection, sparking legal controversies over copyright exceptions and limitations. Looking ahead, resolving the conflicts between GenAI technology and copyright law will require joint efforts from the international community. This article proposes that amending the Berne Convention and the WIPO Copyright Treaty to establish new international standards, create mechanisms for international cooperation, clarify the doctrine of fair use, require transparency, facilitate the functioning of copyright collective management organizations, and develop compensation mechanisms for copyright holders are crucial steps towards balancing innovation and protection. This approach not only calls for up-to-date legal systems but also necessitates collaborative efforts from the technological, industrial, and academic sectors to explore new copyright models that can both incentivize innovation and safeguard copyright holders' rights. As GenAI technology continues to evolve, future considerations must also address issues of technological ethics, model bias, and data privacy, ensuring that technological development aligns with the public interest and avoids exacerbating inequalities. Building a comprehensive regulatory framework that encourages technological innovation while ensuring fairness, transparency, and reliability in its application will be a long-term global challenge.

Funding

This article is funded by the 2021 National Social Science Foundation Project (21BXW040).

Intellectual property

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing, we confirm that we have followed the regulations of our institutions concerning intellectual property.

Research ethics

We further confirm that this article complies with research ethics.

Authorship

All listed authors meet the ICMJE criteria. We attest that all authors contributed significantly to the creation of this manuscript, each having fulfilled criteria as established by the ICMJE. We confirm that the

[244] Geiger and Bulayenko (n 191).

manuscript has been read and approved by all named authors. We confirm that the order of authors listed in the manuscript has been approved by all named authors.

We the undersigned agree with all of the above.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.